

X-rays breaking out of pre-explosion ejecta mark a supernova's first light

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Massive stars die as core-collapse supernovae, whose optical light emerges days after the implosion. Theory predicts that the initial collapse-driven shock, upon breaking through the star and dense circumstellar medium, emits a brief thermal flash of soft X-rays and ultraviolet. Yet these elusive first signals have remained largely undetected, owing to limited wide-field soft X-ray monitoring. Here we report the discovery of a soft X-ray flash, EP260321a, followed days later by a broad-lined supernova from an envelope-stripped progenitor. Its X-ray spectrum, best modeled with blackbody, establishes it as the long-sought archetypal shock breakout. The burst's duration and energetics place the breakout at a radius of 300 solar radii, tracing a dense surrounding shell and revealing abrupt mass ejection within the final month before collapse.

One sentence summary: Einstein Probe detected the archetypal thermal soft X-ray shock breakout of a core-collapse supernova from a pre-explosion shell.

Massive stars end their lives as energetic supernova explosions, which have been discovered and studied invariably via their optical light that emerges days after the core implosion. Yet the physical processes during and before such explosions are poorly understood. This is due largely to the lack of detection of neutrino and electromagnetic signals in the prompt and very early stages. Theory predicts that a neutrino-driven shock wave is launched immediately following the stellar core-collapse. The explosion shock, propagating outward through the dense stellar envelope, is radiation-mediated, wherein photons are effectively trapped, governing momentum and energy transport. At the moment the shock breaks out of the stellar envelope or a dense circumstellar medium (CSM), photons of this trapped radiation escape from the shock front, producing the first light of the supernova. Such shock breakouts (SBO) are predicted to emit a burst of thermal spectrum in the soft X-ray to ultraviolet (UV) band, lasting from tens to thousands of seconds, depending on the size of the enveloping material and total energy output (1, 2, 3). Supernova SBO events provide a direct probe of the explosion energetics, the progenitor stars, and their immediate environment. Furthermore, they offer more precise timing of core-collapse events than optical supernova light does, allowing for more efficient searches for associated neutrinos and possible gravitational-wave signals.

Supernova SBOs are elusive and remain largely undetected, owing to their fleeting and faint nature and a historical lack of sensitive soft X-ray and UV wide-field monitors. To date, despite a few claimed candidates (4, 5), only two X-ray SBO events have been reported, both by *Swift* over the past two decades (6, 7, 8, 9); however, neither matches theoretical predictions: their spectra deviate largely from the simple thermal form because of complex physical processes, and one is associated with a relativistic jet-driven gamma-ray burst. A systematic search for SBOs is now within reach, enabled by the Einstein Probe (EP) (10, 11), which offers a large field of view, excellent soft-X-ray sensitivity, and rapid autonomous follow-up. Here we report the discovery by EP of a fast X-ray transient, followed days later by a broad-lined Type Ic supernova from an envelop-stripped progenitor. Its X-ray emission is soft and matches a single-temperature blackbody, establishes it as a long-sought bona fide supernova SBO.

Discovery of the fast X-ray transient EP260321a

During a survey observation, the Wide-field X-ray Telescope (WXT) onboard the Einstein Probe detected an X-ray transient designated EP260321a (12) on 21 March 2026 at 12:30:18 UTC (T_{det}). An autonomous observation by EP’s Follow-up X-ray Telescope (FXT) was triggered and started from $T_{\text{det}} + 215$ s, further localizing this uncataloged source to a position consistent with the galaxy SDSS J095942.88+002506.2 at a redshift of $z = 0.0344$ (see Ref. (13)). A General Coordinates Network (GCN) Notice was issued automatically at $T_{\text{det}} + 329$ s. Two GCN Circulars were issued at $T_{\text{det}} + 31.4$ min and $T_{\text{det}} + 16$ hr, which identified EP260321a as a supernova shock breakout candidate (14, 15) and induced extensive multi-wavelength follow-up observations. WXT had in fact captured the onset of the transient at $T_0 = 2026-03-21T12:16:08$ ($T_{\text{det}} - 850$ s) in the preceding survey observation. The X-ray light curve increased linearly over ~ 700 s, followed by an exponential decay (see the lower panel of Figure 1). The entire evolution of the decay stage was covered by FXT up to $T_0 + 2300$ s. Generally, the X-ray light curve evolved smoothly, with no significant substructure. Several monitoring observations with FXT were also conducted from $T_0 + 0.4$ days to $T_0 + 20$ days, and the Chandra X-ray Observatory observed EP260321a on $T_0 + 15$ days (16). All these subsequent observations resulted in nondetections. Meanwhile, although the source was covered by several gamma-ray monitors around the time of the outburst, no gamma-ray burst (GRB) counterpart to EP260321a was reported.

The X-ray spectra of WXT (T_0 to $T_0 + 2200$ s) and FXT ($T_0 + 1065$ s to $T_0 + 2359$ s) are well described by an absorbed blackbody function with an intrinsic absorption column density of $N_{\text{H}} = 8.4^{+1.6}_{-1.5} \times 10^{20} \text{ cm}^{-2}$ (in excess of the Galactic absorption of $2.64 \times 10^{20} \text{ cm}^{-2}$) (17) and blackbody temperatures of $kT_{\text{obs}} = 124^{+7}_{-6} \text{ eV}$ and $112.7^{+2.1}_{-2.1} \text{ eV}$ for WXT and FXT, respectively, where k is the Boltzmann constant (see Figure 2). Although a cutoff power-law model also provides a statistically acceptable fit, the best-fitting parameters effectively mimic a blackbody spectrum, whereas a simple power-law model is strongly disfavored (Methods). From the blackbody spectral fits, it can be calculated that the 0.3–10 keV luminosity of EP260321a reached its peak value of $L_{\text{p,X}} = (1.0 \pm 0.3) \times 10^{45} \text{ erg s}^{-1}$ at $t_{\text{p,X}} = 690 \pm 60$ s, which yields an apparent blackbody radius of $(L_{\text{p,X}}/4\pi\sigma T_{\text{obs}}^4)^{1/2} \sim 10 R_{\odot}$. The total radiated energy in the X-ray band (0.3–10 keV) is $E_{\text{th,X}} = (8.7 \pm 0.6) \times 10^{47} \text{ erg}$, with a duration of $T_{90} \sim 1750$ s. Accompanied by a decrease

in flux after the peak, the blackbody temperature decreases slightly from a maximum value of ~ 135 eV to ~ 110 eV. The temperature and total radiated energy of EP260321a indicate that it is the softest and least luminous among all extragalactic fast X-ray transients detected by EP to date (18, 19). The blackbody spectrum, smooth light curve, and extremely low luminosity of EP260321a all suggest that it is a canonical SBO event. In the left panel of Figure 3, we compare the X-ray light curve of EP260321a with two previously reported SBO events: XRF 060218 (6, 7) and XRO 080109 (9, 8), both of which are dominated by a nonthermal component and have lasting tail emission. In comparison, EP260321a was no longer detected after the second orbit of the autonomous FXT observation from $T_0 + 5137$ s to $T_0 + 8111$ s, which may simply result from the absence of the nonthermal component.

Multi-wavelength follow-up and association with SN 2026gzf

Our optical and near-infrared follow-up campaign was carried out using multiple ground-based facilities. Only 40 min after T_0 , the 0.7 m Thai Robotic Telescope began detecting the optical counterpart of EP260321a in the R band, and shortly thereafter the 1.6 m Mephisto telescope in either the ugi or vrz bands (20, 21), showing that the counterpart appears on top of a north-western bright blue knot of the host galaxy SDSS J095942.88+002506.2 (see the upper panel of Figure 1) and that there is only slight flux variations within the first few hours. Approximately one day later, the fluxes across all filters started to increase with a temporal index less than 1, peaking around 12 days, as presented in Figure 4. High-quality spectra were obtained for the optical counterpart of EP260321a, spanning the phase from $T_0 + 2.7$ days to $T_0 + 56.4$ days, which allow us to identify it as a broad-lined Type Ic supernova (SN Ic-BL), SN 2026gzf (see also Refs. (22, 23, 24, 25)) in the galaxy, surprisingly in a local environment of extremely low metallicity. Furthermore, the bolometric light curve of SN 2026gzf shows a close resemblance to that of SN 2006aj (26), SN 1998bw, and SN 2025kg (19), except for its relatively weak presupernova optical bump. In summary, the association of EP260321a with SN 2026gzf provides further support for its identity as a supernova SBO event.

We performed radio observations of EP260321a/SN 2026gzf at multiple frequencies in the range 0.7–23 GHz from $T_0 + 5$ days to $T_0 + 31$ days using MeerKAT, ATCA, e-MERLIN, and uGMRT. We also compiled radio observations from GCN Circulars. No reliable radio counterpart

is detected (Methods). These upper limits lie well below the radio afterglow fluxes of the majority of GRBs (27, 28), disfavoring the existence of an on-axis relativistic jet (Methods; see also Ref. (23)). This is consistent with the nondetection of EP260321a in gamma rays. Furthermore, even for an off-axis jet or a choked jet, higher energy emission than EP260321a could also be expected to arise from the inner cocoon or the cocoon breakout (29, 30, 31). Overall, these radio and gamma-ray constraints strengthen our claim that the soft X-ray emission of EP260321a arises from supernova SBO.

Supernova shock breakout origin of EP260321a

Supernova SBO occurs when the explosion-driven shock reaches a photon diffusion surface at an optical depth of $\tau_{\text{SBO}} \approx c/v_{\text{sh}}$ (the speed ratio of light to shock wave), before which the heat converted from the kinetic energy of the shock cannot diffuse out efficiently, and thus the shock is radiation-mediated. After SBO, the heat trapped behind the shock starts to be released to produce thermal emission in the UV and soft X-ray bands. However, specifically, a radiation-mediated shock moving at a much faster velocity than $v_{\text{sh}} \gtrsim 0.1 c$ would deviate substantially from local thermal equilibrium and lead to a characteristic color temperature much higher than 0.1 keV (see Ref. (32)). Therefore, the observed thermal spectra and relatively low temperature of EP260321a strongly suggest that the shock is nonrelativistic, with a velocity $v_{\text{sh}} \lesssim 0.1 c$. Meanwhile, the shock velocity is also required to be not much lower than $\sim 0.1 c$, such that the kinetic energy of the supernova ejecta moving faster than v_{sh} can be matched to the observed X-ray energy.

According to the peak time of EP260321a, the breakout radius r_{SBO} can be inferred to be around $300 R_{\odot}$, from either photon diffusion or the light-travel-time effect (Methods). This is much larger than the progenitor's radius, as it should be a compact Wolf-Rayet star for SNe Ic-BL (i.e., $r_* \lesssim R_{\odot}$) (33). Therefore, the EP260321a SBO event is very unlikely to occur at the stellar surface and, instead, is more likely to be driven by the interaction of the supernova ejecta with dense and optically thick CSM. More notably, r_{SBO} is also considerably larger than the apparent blackbody radius directly inferred from observations. The primary reason for this difference is that the observed spectra, which appear as blackbody, are actually Comptonized spectra determined by a scattering equilibrium between electrons and photons, due to the high speed of the shock. In this case, the

effective temperature T_{th} defined from the Stefan-Boltzmann law for the radiation energy density as $u = aT_{\text{th}}^4$ is a few times lower than the color temperature T_{obs} of the observed spectra (32, 34), where u is the energy density behind the radiation-mediated shock and a is the radiation constant. Furthermore, the light-propagation effect would also cause the SBO emission at a given time to be dominated by a ring-like region instead of the entire sphere.

Since the post-shock energy density can also be related to its upstream kinetic energy density by $u = (6/7)\rho_{\text{SBO}}v_{\text{sh}}^2$, we can derive the density of the upstream CSM to be $\rho_{\text{SBO}} = 7aT_{\text{obs}}^4/(18f_{\text{col}}^4v_{\text{sh}}^2) \sim 10^{-11} \text{ g cm}^{-3}$, where the color correction factor $f_{\text{col}} \equiv T_{\text{obs}}/T_{\text{th}} \sim 3.5$ is introduced to represent the deviation from complete local thermal equilibrium. Then, according to the SBO condition as $\tau_{\text{SBO}} \approx \kappa_X \rho_{\text{SBO}}(r_{\text{out}} - r_{\text{SBO}}) \approx c/v_{\text{sh}}$, we find that the outer radius of the CSM is only slightly larger than the breakout radius as $(r_{\text{out}} - r_{\text{SBO}}) \sim 50 R_{\odot}$, where $\kappa_X = 0.2 \text{ cm}^2 \text{ g}^{-1}$ is the opacity of fully ionized He/C/O-rich material for soft X-rays. Furthermore, being constrained by the energy budget of X-ray emission, the total mass of the shocked CSM that contributes to the X-ray emission can be derived as $M_{\text{CSM}} \sim 10^{-4} M_{\odot}$ from $M_{\text{CSM}}v_{\text{sh}}^2/2 \sim E_{\text{th,X}}$. The constraints on M_{CSM} as well as ρ_{SBO} indicate that the shocked CSM is accumulated by the shock from a narrow range around the breakout radius with a width of $\Delta r \approx M_{\text{CSM}}/(4\pi r_{\text{SBO}}^2 \rho_{\text{SBO}}) \sim 50 R_{\odot}$, which is comparable to the value of $(r_{\text{out}} - r_{\text{SBO}})$. Thus, we suggest that the SBO emission does not originate from the CSM at radii much smaller than r_{SBO} . This implies that the CSM is most likely confined within a thin shell, as any material located at smaller radii would be compressed to near r_{SBO} if present.

Detailed modeling of the X-ray light curve and temperature evolution of EP260321a is presented by the solid line in the lower panel of Figure 1, where a confined CSM shell with a truncated wind density profile is adopted for $r_{\text{in}} \approx 260 R_{\odot}$ and $r_{\text{out}} \approx 320 R_{\odot}$ and the shock velocity is set at $v_{\text{sh}} \approx 0.089 c$. Here, it is considered that all heat accumulated by the shock contributed to the SBO X-ray emission. The most promising origin of such a shell-like CSM is an episodic mass ejection by the progenitor star before the supernova explosion (22), which could be a wave-driven outburst and powered by late-stage nuclear burning (35, 36, 37, 38, 39). The inferred CSM mass and radius of EP260321a also appear to be broadly consistent with the typical properties expected from wave-driven outbursts (37). The corresponding final mass ejection likely occurred at $t_{\text{CSM}} \approx r_{\text{SBO}}/v_{\text{CSM}} \lesssim 20$ days before the supernova explosion, where the velocity of wave-driven

outbursts for a stripped-envelope star is taken as a few 100 km s^{-1} (36, 37), suggesting a possible connection to oxygen/neon burning inside the progenitor. In principle, such episodic wave-driven outbursts can take place multiple times for a stripped-envelope star, thereby forming a series of distinct CSM shells at different radii. However, assuming that the mass losses during each ejection are comparable, the outer CSM shells at larger radii are likely optically thin, and thus the interaction of the supernova ejecta with these outer shells could not produce a significant observational signal.

The optical supernova SN 2026gzf

As a baseline test, we fit the multiband light curves of SN 2026gzf with a purely radioactive model powered by the decay chain of ^{56}Ni using `TransFit`, a time-dependent radiative-diffusion code for supernova light-curve inference (40). The best-fit model, shown in Figure 4, requires a ^{56}Ni mass of $0.45 M_{\odot}$ and an ejecta mass of $1.85 M_{\odot}$. To reproduce the shallow rise and the ~ 10 -day peak with radioactive heating alone, the model further requires nearly complete ^{56}Ni mixing, with $x_{\text{Ni}} = 0.993$, which corresponds to an almost homogeneous distribution of radioactive material throughout the ejecta. Such strong mixing could in principle be enhanced by explosion-driven hydrodynamic instabilities in a compact Wolf–Rayet progenitor that experienced vigorous late-stage mass loss (41, 42, 43). However, this level of ^{56}Ni homogenization is difficult to obtain in multidimensional explosion simulations, which generally produce asymmetric, clumpy, or finger-like distributions of iron-group material rather than a nearly uniform radioactive component (41, 42). If weaker mixing is imposed, the early emission can be reproduced only by reducing the ejecta mass to below $\sim 1 M_{\odot}$, which is unusually low for a stripped-envelope supernova (44, 45). The relatively high nickel-to-ejecta mass ratio ($\gtrsim 25\%$) suggests that other energy sources may play an important role in powering SN 2026gzf (Methods).

Event rates and implications

Finally, from the single detection of EP260321a in the systematic survey with EP-WXT, we derive a local event rate density of X-ray SBO transients of $\rho_{0,\text{SBO}}(\text{EP260321a}) \geq 170 \text{ Gpc}^{-3} \text{ yr}^{-1}$ above a peak luminosity of $1.0 \times 10^{45} \text{ erg s}^{-1}$ (see Methods). This event-rate density sits between the more

common XRO 080109, associated with an SN Ib/c, and the more luminous and rare event XRF 060218, associated with an SN Ic-BL, forming a power-law luminosity function for X-ray-detected SBOs with an index of 1.3 ± 0.4 as shown in the right panel of Figure 3. This fact, together with more common SBOs observed in the optical band in Type II SNe (46, 47, 48), implies a unified scheme for stellar shock breakouts, with rarer, more luminous events arising from more extreme stellar explosions involving higher angular momentum and more envelope stripping. Furthermore, the detection of supernova SBOs yields more precise timing of core-collapse events compared to optical supernovae. This allows for more efficient searches for associated neutrinos and potential gravitational-wave signals in the nearby universe. Future continued observations with Einstein Probe and other wide-field monitors, together with simultaneous multi-messenger observations in gravitational waves and neutrinos (49) at the time of SBOs, will further constrain SBO physics and may uncover how massive stars end their lives.

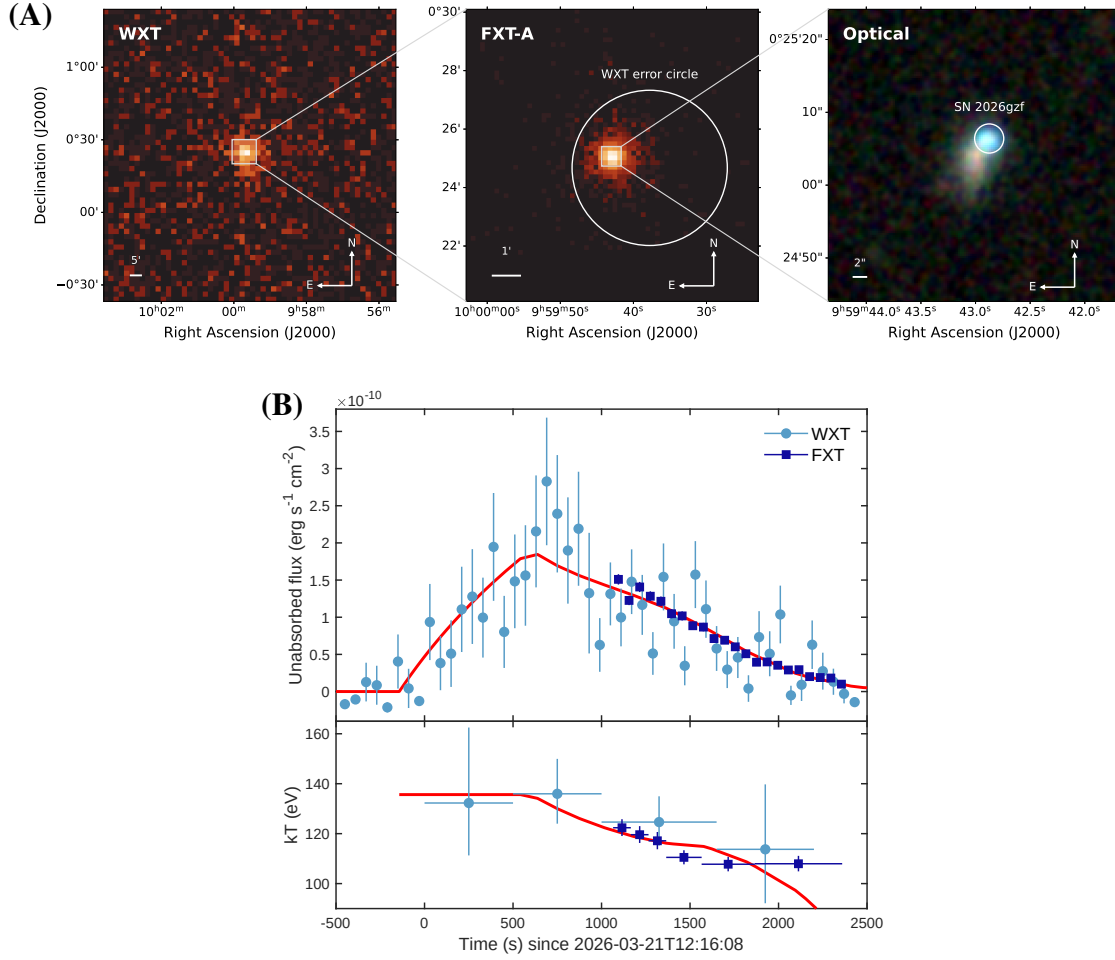


Figure 1: X-ray/optical images and X-ray evolution of EP260321a. (A) WXT and FXT-A image in 0.4–2.0 keV. The white circle shows the positional error circle of WXT with a radius of $2.7'$ (90% confidence level). The optical image is the composite Mephisto-*gri* band images taken at $\sim T_0 + 2.3$ hr. (B) The evolution of unabsorbed X-ray flux in 0.4–2.0 keV (upper) and temperature (lower) under the absorbed blackbody model. The transient began at $T_0 = 2026-03-21T12:16:08$ UTC (corresponding to the time of a 3σ excess above the background). Raw count rates were converted to the unabsorbed flux using the factors determined by the time-resolved spectral fitting (Methods). The vertical error bars show the 1σ uncertainty and the horizontal error bars indicate the corresponding time intervals. The solid lines represent the best fit with a supernova SBO model (Methods).

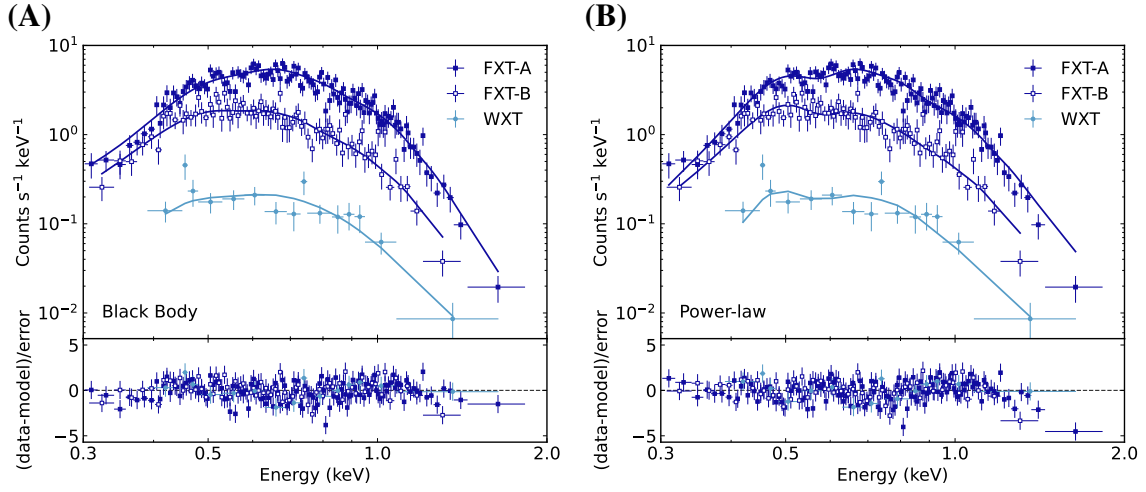


Figure 2: X-ray spectra of EP260321a. The WXT spectrum includes data from T_0 to $T_0 + 2200$ s; the FXT spectra include data from $T_0 + 1065$ s to $T_0 + 2359$ s. The count rate of the FXT-B spectrum is significantly lower than that of FXT-A, since the counts in the core of the point-spread function of FXT-B were excluded in the spectrum to mitigate pile-up effect (Methods). The data points are shown with 1σ uncertainty and the lines represent the best-fit absorbed blackbody (A) and power-law (B) models.

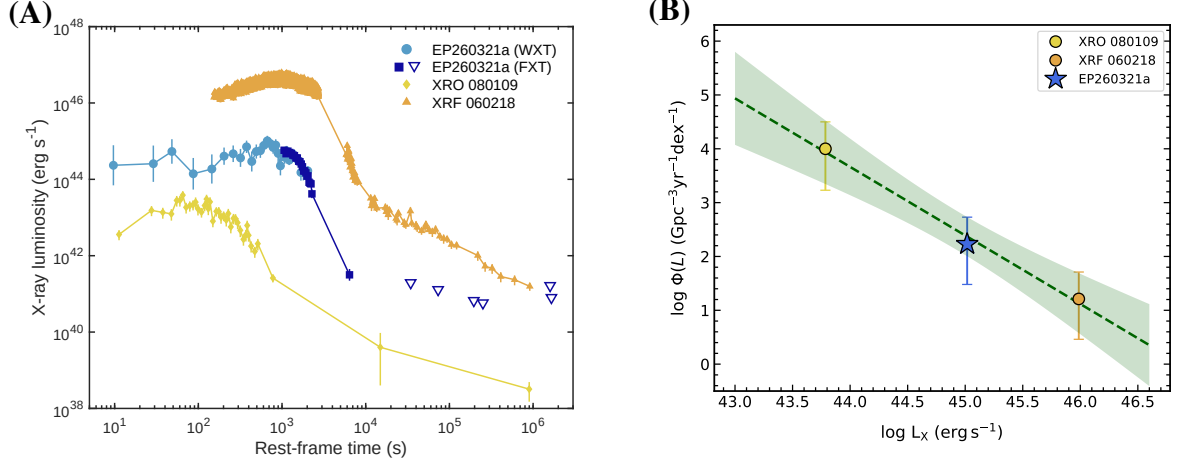


Figure 3: X-ray light curves and luminosity function of SN SBOs. (A) X-ray luminosity light curves of SN SBO events, including EP260321a, XRF 060218 (6, 7), and XRO 080109 (9, 8). The data points are shown with 1σ uncertainty. The open triangles represent the upper limits derived from the FXT monitoring observations of EP260321a. The data of XRF 060218 are obtained from the *Swift* X-ray telescope light-curve repository (50, 51). The data of XRO 080109 are obtained from Ref. (52). (B) X-ray luminosity function of SN SBOs. The best-fit result from a power-law model and its 1σ uncertainty are shown as a dashed green line and the light-green shaded region, respectively. The peak luminosity of the thermal component of XRO 060218 is adopted (6). All luminosities are in the 0.3–10 keV band.

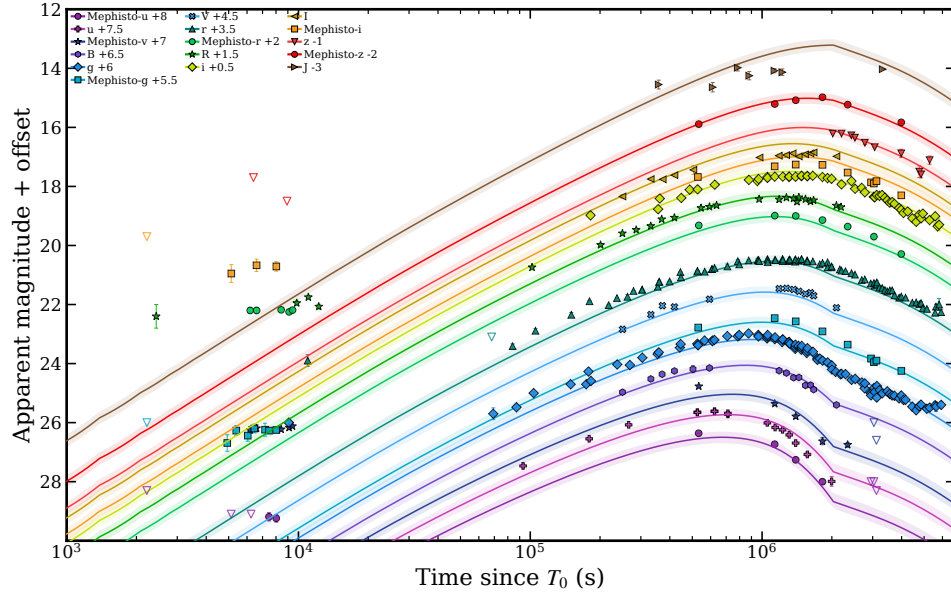


Figure 4: Multiband optical and near-infrared evolution of SN 2026gzf. Apparent magnitudes are plotted as a function of time, with vertical offsets applied to individual filters for clarity. Open downward triangles mark upper limits. The solid lines give a theoretical fit of the light curves by invoking power due to the radioactive decay of ^{56}Ni , with the shaded bands representing the 1σ model uncertainty.

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Author contributions: W.Y. has been serving as Principal Investigator of the Einstein Probe project since the mission proposal phase. W.Y., X.-F. W., and B.Z. initiated and coordinated this study and led the discussions.

Y.L., H.-Q.C., and R.-D.L. processed and analysed the WXT data. Y.L., C.-K.L., Y.-H.Y., and R.-D.L. processed and analysed the FXT data. H.-Q.C., H.-W.P., and Y.L. contributed to the development of WXT data analysis software and its calibration. S.-M.J. and C.-K.L. contributed to the development of FXT data analysis software and its calibration. H.S. and J.-W.H. contributed to the estimate of the event rate. Q.-J.H., Z.-C.Z., D.-Y.L., and X.M. were the transient advocates on 21 March 2026, and contributed to the discovery and preliminary analysis of the event.

D.X. coordinated the optical data reduction and analysis. D. X., X. L., J. F., Z.-P. Z., K. C., S.-Y. F., L. B.-H., S.-Q. J., W.-X. L., K. N., S. T., and Z. F. contributed to the TRT, ALT, and NOT observations. G.-W. D., Q.-R. W., Y. F., X.-K. L., X.-W. L. and Y. P. contributed to the Mephisto observations. J.-A. J, M.-X. C, Z.-Q. J, and Z.-L. X. contributed to the WFST observations. W.-K.Z, A.F., and T.B. contributed KAIT and Shane observations. A.S.P., N.S.P., A.V.V., I.A.Z., A.S.M., E.V.K., M.V.E., O.S., V.G., and V.V. contributed to the Russian multiple telescopes. R.-D.L and X. F contributed to the WHUT observations. B.W., R.-Z. Li, and J.M. contributed to the GMG observations. J.-J.J., Z.F., H.W., H.-Y. M., Y. Z., and M.H. contributed to the Xinglong-2.16m and XL-80 observations. Y.Y. contributed to the VLT, KAIT, and Shane observations.

T.A. coordinated the radio observations. T.A., A.L. W., X.F.W., J.J.G., Y. Q.L., S.B. Z., X.Y. W, C.Y. D. contributed to the ATCA observations. A.-L.W., T.A., X.-F.W., J.-J.G., X.-Y.W., C.-Y.D., and Z.-W.L. contributed to the MeerKAT observations. G. G., L. P., G.B., and A.-L.T. contributed to the e-MERLIN observations. A.B., D.E., A.B., V.B., G.C. A., D.K.S., and A.P.S. contributed to the uGMRT observations.

Y.-W.Y., J.-P.Z., and Z.L. led the theoretical investigation of the event. Y.-H.Z., G.-L.W., L.-D.L.,

K.-R.N., J.-S.Z., and B.W. contributed to the theoretical investigation of the event.

H-W. P, C. B, Z.-M. C, J.-X. H, J.-W. H, S.-M. J, Z.-X. L, S.-L. S, C. Z, X.-J. S, Y.-H. Z, Y.-H. C, C.-Z. C, J. G, D.-W. H, M.-H. H, G. J, C.-K. L, C. Z. L, H.-Q. L, L.-M. S, H.-Y. T, H.-T. X., and C.-B. X contributed to the hardware and software development and the science operations of the Einstein Probe mission. J.-N, K. N, S.-N. Z, B. C, H. F, C.-C. J, P. B, A. R, N. R, and J. S contributed to the science management and collaboration of the Einstein Probe project.

All authors joined various kinds of discussions at different stages of the work.

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Supplementary materials

Materials and Methods

Figures S1 to S4

Tables S1 to S5

References (51-106)

Supplementary Materials for

X-rays breaking out of pre-explosion ejecta mark a supernova's first light

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This PDF file includes:

Materials and Methods

Figures S1 to S4

Tables S1 to S5

Materials and Methods

Einstein Probe (EP) observations and data reduction

EP260321a triggered the Wide-field X-ray Telescope (WXT) onboard processing unit at 2026-03-21T12:30:18 UTC during a survey observation (ObsID 11900654211) with a signal-to-noise ratio of 7.0. This source was also detected in the preceding survey observation (ObsID 11900654465) and the autonomous follow-up observation (ObsID 01709259023). Thus, WXT observations covered the entire onset and decay phase of this transient with a total exposure time of ~ 3100 s.

The X-ray events were processed and calibrated using the dedicated data-reduction software and calibration database (CALDB) developed for the WXT instrument (Liu et al., in prep.). The CALDB was constructed based on the results of ground calibration experiments (53), following procedures previously applied to a WXT prototype (54). Photon positions were transformed into celestial coordinates in the J2000 reference frame. The energy of each event was determined using the bias and gain parameters provided in the CALDB. After identifying and flagging bad and flaring pixels, single-, double-, triple-, and quadruple-pixel events without anomalous flags were retained to produce the cleaned event file.

The 0.4–2.0 keV image was generated from the cleaned event list (Figure 1). Source and background light curves, as well as time-resolved spectra, were extracted from a circular region with a radius of $9'$ and an annular region with inner and outer radii of $18'$ and $36'$, respectively. The peak count rate in the source region was only 0.01 counts per frame, indicating that the pile-up effect in the WXT data was negligible.

An autonomous follow-up observation was carried out with the Follow-up X-ray Telescope (FXT) (55) and lasted for two orbits with an exposure time of 4267 s. The FXT-A and FXT-B were working in the partial-window and full-frame modes, respectively. The Level 1 event files were processed using the FXT data-analysis software (v1.30) together with the corresponding CALDB (v1.30). The reduction procedure included particle-event identification, pulse-invariant (PI) conversion, grade assignment and selection ($\text{grade} \leq 12$), flagging of bad and hot pixels, and the screening of good time intervals based on housekeeping data. This pipeline produced cleaned event files. For FXT-A, source and background light curves, as well as time-resolved spectra, were extracted from a circular region with a radius of $1'$ and from two adjacent circular regions, each

with a radius of $3'$, respectively. For FXT-B, the source region was an annulus with inner and outer radii of $20''$ and $1'$ to mitigate pile-up in the full-frame mode. The corresponding response files were then generated with the point-spread function correction. In addition to the autonomous follow-up observation, FXT conducted six observations to monitor the long-term X-ray behavior of EP260321a (Extended Data Table S1). No significant detection was found in these observations.

X-ray spectral analysis

Since there is no significant signal above 2.0 keV, the X-ray spectral analysis was performed over 0.4–2.0 keV and 0.3–2.0 keV for WXT and FXT in XSPEC, respectively. The spectra of FXT-A and FXT-B were generated separately and fitted simultaneously. The W-statistic (*cstat* in XSPEC) (56) was employed in deriving the best-fit spectral parameters. The integrated and time-resolved spectra were rebinned to at least 10 and 3 counts per bin, respectively. The integrated spectra of the WXT ($[T_0, T_0 + 2200 \text{ s}]$) and FXT ($[T_0 + 1065 \text{ s}, T_0 + 2359 \text{ s}]$, i.e., the first orbit of the autonomous follow-up observation) can be well described by an absorbed blackbody model *tbabs*ztbabs*zbody*, where the first and second components respectively account for the Galactic absorption $N_{\text{H, Gal}}$ and the intrinsic absorption N_{H} , and the third component is a blackbody function in the source rest frame. The column density of the Galactic absorption in the direction of EP260321a was fixed at $2.64 \times 10^{20} \text{ cm}^{-2}$ (Ref. (17)) and the redshift was fixed at 0.0344. If the N_{H} of FXT is linked with that of WXT, the best-fit parameters are $N_{\text{H}} = 8.4_{-1.5}^{+1.6} \times 10^{20} \text{ cm}^{-2}$, $kT_{\text{obs}} = 124_{-6}^{+7} \text{ eV}$ (WXT), and $112.7_{-2.1}^{+2.1} \text{ eV}$ (FXT) with an acceptable statistic $\text{W-stat}/(\text{d.o.f.}) = 245.7/231$. The goodness of fit is not significantly improved with independent N_{H} . If the integrated spectrum was fitted by an absorbed power-law model *tbabs*ztbabs*zpowerlaw*, the goodness of fit is much poorer with a statistic of $\text{W-stat}/(\text{d.o.f.}) = 272.2/231$, and the best-fit photon index (~ 10) is also unphysical (right panel of Figure 2). The difference of the Bayesian Information Criterion (57) ($\Delta BIC = 26.5$) indicates strong evidence against the absorbed power-law model. A cutoff power law could fit the integrated spectrum acceptably with a cutoff energy of $\sim 120 \text{ eV}$ and a photon index of ~ -2 , which effectively mimics a blackbody function.

To investigate the possible spectral evolution, the WXT observation was divided into four segments; the FXT observation during the first orbit of the autonomous follow-up observation was divided into six segments of each FXT module (12 FXT segments in total). The resulting 16

spectra were simultaneously fitted by an absorbed blackbody model $tbabs*ztbabs*zbbody$. The N_H values of all segments were linked assuming no significant variation of the intrinsic absorption. The blackbody temperatures of FXT-A and FXT-B in the same segment were also linked. The spectra of all segments are consistent with the absorbed blackbody model. The best-fit N_H is $7.9^{+1.7}_{-1.6} \times 10^{20} \text{ cm}^{-2}$. The blackbody temperature slightly decreased from $\sim 135 \text{ eV}$ to $\sim 110 \text{ eV}$ (Extended Table S2; Extended Data Figure S1).

In the second orbit of the autonomous follow-up observation ($[T_0 + 5137 \text{ s}, T_0 + 8111 \text{ s}]$), EP260321a was still well detected with a significance of ~ 5 . The flux was estimated to be $(7.1 \pm 2.5) \times 10^{-14} \text{ erg cm}^{-2} \text{ s}^{-1}$ in 0.4–2.0 keV by assuming an absorbed blackbody with $N_H = 8 \times 10^{20} \text{ cm}^{-2}$ and $kT_{\text{obs}} = 100 \text{ eV}$. Additionally, the 3σ upper limits for the nondetection in the other FXT observations were derived under the same spectral shape (Extended Table S1).

Optical and near-infrared photometric observations

Our ground-based optical and near-infrared photometric observations were conducted using multiple facilities and are still ongoing. The current results are summarized in Extended Table S3. The 1.6 m Mephisto telescope began imaging the field 36 min after T_0 , operating simultaneously in either the *ugi* or *vrz* bands (58, 59). The Jinshan project commenced continuous monitoring in the *grizJ* bands using the 1.0 m 100A/100B/100C and 0.5 m 50A telescopes starting at $T_0 + 1.8 \text{ hr}$. The 0.76 m Katzman Automatic Imaging Telescope (KAIT; Filippenko et al. 2001) observed the field in the *BVRI* and *Clear* bands from $T_0 + 16.5 \text{ hr}$. Further optical observations were carried out with the 2.5 m Wide Field Survey Telescope (WFST) in the *ugr* bands starting at $T_0 + 2.04 \text{ days}$. Additional photometric data were obtained with the Tsinghua-NAOC 0.8 m telescope (TNT) (60), the 0.7 m telescope of the Thai Robotic Telescope network (TRT) located at the Spring Brook Remote Observatory (SBO) and the Cerro Tololo Inter-American Observatory (CTO), the SAO RAS 1.0 m Zeiss-1000 Optical Telescope (Zeiss-1000) and the 0.5 m AS-500/2 telescope, the 2.56 m Nordic Optical Telescope (NOT), the 2.6 m Shajn Telescope (ZTSh) and the 50-cm Maksutov telescope (MTM-500) of the Crimean Astrophysical Observatory, and the 1.5 m AZT-33IK telescope of the Mondy Observatory. Along the line of sight of EP260321a/SN 2026gzf, the Galactic extinction is $E_{B-V} = 0.021 \text{ mag}$ (61).

All photometric images were calibrated following standard procedures, including bias subtrac-

tion, dark-current subtraction (for CMOS detectors), and flat-field correction. We used *astrometry.net* (62) and SCAMP (63) for astrometric calibration, IRAF (64) and SWarp (65) for image stacking. Aperture photometry on the reduced images was carried out with SExtractor (66). Mephisto *uvgriz* photometry was calibrated against the synthetic photometry derived from the recalibrated Gaia BP/RP (XP) spectra (67, 68). The Sloan *griz* (as well as $g'r'i'z'$) photometry was calibrated against nearby reference stars from the Pan-STARRS DR2 catalog in the AB magnitude system (69). For the Johnson-Cousins filters, the *BVRcIc*-band photometry in the Vega system was calibrated using magnitudes converted from the Sloan system (<https://www.sdss.org/dr12/algorithms/sdssUBVRITransform/#Lupton>). To subtract the host contribution, we used HOTPANTS (70) to perform image subtraction. As template images we used multiband stack images from PanSTARRS-1 (PS1) survey, unfiltered coadded images from the Catalina Sky Survey (71), *BV*-band archival images from the Las Cumbres Observatory (LCO) network (72), and a *J*-band archival image from the UKIRT Infrared Deep Sky Survey (73) (UKIDSS). Images from Mephisto and WFST were subtracted using their own archival templates. Photutils (74) was then used to perform aperture photometry on the difference image to obtain the transient magnitude.

Optical spectroscopic observations

To investigate the ejecta properties and chemical evolution of SN 2026gzf, we conducted a multi-epoch spectroscopic campaign spanning from $T_0 + 2.7$ days to $T_0 + 56.4$ days. 19 high-quality spectra were obtained, with their detailed observational parameters listed in the Extended Table S4. The resulting spectral sequence is illustrated in the left panel of Extended Data Figure S2, providing a comprehensive view of the transient’s spectral transformation. All spectroscopic data were reduced using standard procedures within IRAF (64, 75), including bias subtraction, flat-fielding, and optimal extraction. Wavelength and flux calibrations were performed using daily HeNe comparison-lamp exposures and spectrophotometric standard stars, respectively.

Low-resolution optical spectra were acquired using the ALFOSC spectrograph on the 2.56 m Nordic Optical Telescope (NOT) between 2026 March 26 and May 16; 13 reliable spectra were obtained by the following procedures. We employed Grism #4 in conjunction with a 1.3'' slit, yielding a typical spectral resolution of $R \equiv \lambda/\Delta\lambda \approx 300$ and a usable wavelength coverage of

3800–9200 Å. We performed multiple successive exposures at each epoch, allowing for robust removal of cosmic rays during the image-stacking process. The spectroscopic data were reduced using standard IRAF-based pipelines, including bias subtraction, flat-field correction, and one-dimensional spectral extraction. Wavelength calibration was anchored to HeNe comparison-lamp spectra. Flux calibration was achieved using spectrophotometric standard stars (e.g., Feige 34 or HD 84937) observed during the same nights at similar airmasses, with their reference spectrum taken from the Oke catalog (76).

Follow-up spectroscopy was also conducted using the Kast double spectrograph mounted on the 3 m Shane telescope at Lick Observatory; 3 spectra were acquired between 2026 March 24 and 2026 March 27. The observations utilized a dual-beam configuration with the d55 dichroic mirror, typically employing a 600/4310 grating for the blue channel and a 300/7500 grating for the red channel. This setup provided broad and continuous wavelength coverage from ~ 3500 Å to 10,500 Å. The long 2''-wide slit was aligned at the parallactic angle (Filippenko 1982) to minimize differential light losses caused by atmospheric dispersion. A complete log of the Kast observations, including exposure times and specific instrumental setups for each epoch, is detailed in Extended Table S4.

Additional optical spectra were obtained using the BFOSC instrument mounted on the Xing-long 2.16 m telescope of the National Astronomical Observatories, Chinese Academy of Sciences (NAOC) (77). These observations were carried out between 2026 March 26 and April 15, yielding two high-quality spectra. We utilized the Grism 4 (G4), which, combined with a 2.3'' slit, provided wavelength coverage of 3600–9000 Å. A detailed journal of the BFOSC observations is included in Extended Table S4.

A high-signal-to-noise ratio optical spectrum was obtained on 2026 April 6 using the FORS2 (Focal Reducer and low dispersion Spectrograph 2) instrument mounted on the 8.2 m Very Large Telescope (VLT) UT1 (Antu) at ESO’s Paranal Observatory (78). We utilized the 300V grism in combination with a 1.0'' slit, providing wavelength range of 3410–9330 Å. Observations were carried out in the Polarimetric Multi-Object Spectroscopy (PMOS) mode; 8 integrations were obtained, each with an exposure time of 480 s. We preprocessed the raw data collected at each retarder plate angle and extracted the ordinary and extraordinary beams using standard methods in IRAF. Wavelength calibration was conducted with a root-mean-square accuracy of ~ 0.2 Å. Details

of the FORS2/PMOS data reduction and the derivation of the Stokes parameters can be found in the FORS2 Spectropolarimetry Cookbook and Reflex Tutorial¹ and Refs (79, 80), following the procedures described in Ref (81).

It is clear that there are not [N II] emission lines in the spectral sequence, while prominent Balmer and He I emission lines are present, which reveals that SN 2026gzf is located at a place of extremely low metallicity. Interestingly, the survey images from PanSTARRS and Legacy Survey show that the location of SN 2026gzf is on top of a bright blue knot of the host galaxy, indicating that the SN 2026gzf location is actually very close to highly star-forming regions.

Supernova bolometric light curve

A bolometric light curve for the supernova was constructed using NOT and Shane spectra, combined with multiband photometry from ALT (g, r, i, z, J bands) and Mephisto (u, v, z bands), utilizing temporal interpolation where necessary. Spectral fluxes were scaled to match the multiband photometry. When available, these were complemented by effective fluxes at 3460 Å from the Mephisto u band and those at 12,200 Å from the ALT J band. Before integration, the spectra were extrapolated to assumed zero-flux points at 3000 Å and 13,500 Å. By comparing the spectral energy distribution (SED) of the multiband photometry before and after host-galaxy subtraction, we estimated a host contribution of $(2.5 \pm 0.7) \times 10^{-13} \text{ erg s}^{-1} \text{ cm}^{-2}$, which represents approximately 8%–20% of the total flux depending on the epoch. This contribution was then subtracted from the spectrum-integrated bolometric fluxes. Galactic and host-galaxy reddening were not taken into account, as their contributions are negligible for this supernova.

The results were converted to absolute magnitudes by adopting a distance modulus of $\mu = 35.84$ mag. To account for minor epoch differences between the spectroscopic and photometric data, effective epochs and corresponding uncertainties were assigned to each bolometric point. Magnitude uncertainties include contributions from photometric measurements, host-galaxy subtraction, and spectral flux scaling (fixed at 5%). Spectra from days +2.7 and +4.7 were adopted for epochs +2.2 and +4.2 days, respectively. Similarly, Mephisto photometry from day +6.2 was utilized for days +4.2 and +5.2. While (quasi-)bolometric magnitudes derived from host-subtracted multiband

¹<ftp://ftp.eso.org/pub/dfs/pipelines/instruments/fors/fors-pmos-reflex-tutorial-1.3.pdf>

photometric SEDs generally underestimate the spectrum-integrated results by 0–0.15 mag across most epochs, the discrepancy reaches ~ 0.3 mag at days +2.1 and +2.7. We conservatively adopted mean values for these two earliest epochs, with assigned uncertainties of 0.15 mag. The final bolometric light curve, converted to the supernova rest frame, is shown in the right panel of Extended Data Figure S2.

SBO from a CSM shell

The peak timescale of an SBO signal is determined by both photon diffusion and the light-travel time. If the peak is dominated by photon diffusion, the peak luminosity is comparable to the instantaneous injected luminosity at the SBO, $L_{p,X} \approx 2\pi r_{\text{SBO}}^2 \rho_{\text{CSM}} v_{\text{sh}}^3$. The diffusion timescale $t_{\text{diff}} \approx \kappa_X \rho_{\text{CSM}} \Delta r^2 / c$ is approximately equal to the dynamical timescale $t_{\text{dyn}} \approx \Delta r / v_{\text{sh}}$, both of order $t_{p,X}$, where κ_X is the opacity, Δr is the width of the material outside the SBO radius between the SBO radius and outer boundary, and c is the speed of light. Thus, the SBO radius can be expressed as $r_{\text{SBO}} \approx \sqrt{\kappa_X L_p t_{p,X} / (2\pi c v_{\text{sh}})}$. As described in the main text, the spectrum of EP260321a suggests thermal-like emission, implying that the shock cannot be far from local thermal equilibrium, with $v_{\text{sh}} \lesssim 0.1 c$. Furthermore, as the SBO emission is powered by the conversion of the kinetic energy of the outer ejecta, the inferred radiated energy places another constraint on the shock velocity. For a typical broken-power-law structure of supernova ejecta (82), the available kinetic energy within the outer ejecta above velocity v_{sh} can be expressed as $E_{\text{kin}}(> v_{\text{sh}}) \propto E_{\text{kin}}(v_{\text{sh}}/v_{\text{tr}})^{-5}$, where v_{tr} is the transition velocity. By equating $E_{\text{kin}}(> v_{\text{sh}})$ to $\sim E_{\text{th},X}$, we obtain

$$v_{\text{sh}} \sim 0.09 c \left(\frac{E_{\text{kin}}}{10^{51} \text{ erg}} \right)^{7/10} \left(\frac{M_{\text{ej}}}{1 M_{\odot}} \right)^{-1/2} \left(\frac{E_{\text{th},X}}{10^{48} \text{ erg}} \right)^{-1/5}, \quad (\text{S1})$$

where M_{ej} and E_{kin} are the total mass and kinetic energy of the supernova ejecta, respectively. This implies that the shock velocity cannot be very low, even allowing for reasonable variations in E_{kin} and M_{ej} . Then, the SBO radius determined by photon diffusion can be estimated as

$$r_{\text{SBO}} \approx \sqrt{\frac{\kappa_X L_p t_{p,X}}{2\pi c v_{\text{sh}}}} \sim 270 R_{\odot} \left(\frac{\kappa_X}{0.2 \text{ cm}^2 \text{ g}^{-1}} \right)^{1/2} \left(\frac{L_p}{10^{45} \text{ erg s}^{-1}} \right)^{1/2} \left(\frac{t_{p,X}}{700 \text{ s}} \right)^{1/2} \left(\frac{v_{\text{sh}}}{0.09 c} \right)^{-1/2}. \quad (\text{S2})$$

Alternatively, if the observed peak timescale of EP260321a is attributed to the light-travel effect, the corresponding SBO radius is inferred to be

$$r_{\text{SBO}} \lesssim r_{\text{out}} \approx c t_{p,X} \sim 300 R_{\odot} \left(\frac{t_{p,X}}{700 \text{ s}} \right). \quad (\text{S3})$$

Thus, both the light-travel effect and photon diffusion constrain the SBO radius of EP260321a to be $\sim 300 R_\odot$.

For the light-curve and temperature modeling, we then consider a radiation-mediated shock propagating with a velocity of v_{sh} through a CSM shell confined in a range of $r_{\text{in}} \lesssim r \lesssim r_{\text{out}}$, with a wind-like density profile $\rho_{\text{CSM}}(r) = Kr^{-2}$. The normalization coefficient of the density profile is given by $K = M_{\text{CSM}}/[4\pi(r_{\text{out}} - r_{\text{in}})]$, with M_{CSM} representing the CSM shell mass. At the moment of SBO, where the time zero ($t = 0$) is defined, the optical depth outside the SBO radius satisfies

$$\tau(t = 0) \approx \int_{r_{\text{SBO}}}^{r_{\text{out}}} \kappa_X \rho_{\text{CSM}} dr = \kappa_X K \left(\frac{1}{r_{\text{SBO}}} - \frac{1}{r_{\text{out}}} \right) \approx \frac{c}{v_{\text{sh}}}, \quad (\text{S4})$$

which is contributed by the unshocked CSM. As the shock propagates outward, photons trapped in the inner shocked CSM can gradually diffuse out, thus producing cooling emission (34, 83). We define the layer of the diffusing shocked CSM with optical depth $\tau \approx c/v_{\text{sh}}$ as the *breakout shell*. Then, we calculate the mass M_{diff} of the diffusing shocked CSM and the unshocked CSM (if the shock has not yet swept up all the CSM) outside the breakout shell at different times t . The diffusion of photons for the entire shocked CSM can be determined by $M_{\text{diff}} \approx M_{\text{CSM}}$, thus obtaining a characteristic diffusion timescale $t_{\text{diff,c}}$ for the entire shocked CSM.

The initial energy of the breakout shell at time t is given by $E_{\text{int},0} \approx M_{\text{diff}} v_{\text{sh}}^2 / 2$. Since the CSM is very thin, the internal energy can be approximately treated as constant, i.e., $E_{\text{int}} \approx E_{\text{int},0}$, when $r_{\text{sh}} \lesssim r_{\text{out}}$. Once the shocked material expands beyond the outer edge of the CSM ($r_{\text{sh}} \gtrsim r_{\text{out}}$), adiabatic expansion leads to an evolution of the internal energy as $E_{\text{int}} \propto r^{-1}$. The internal energy can therefore be estimated as $E_{\text{int}} \approx M_{\text{diff}} v_{\text{sh}}^2 r_{\text{out}} / (2r_{\text{sh}})$. Then, we can model the source luminosity lightcurve of the cooling emission as

$$L \approx \begin{cases} \frac{E_{\text{int}}}{t + (r_{\text{out}} - r_{\text{SBO}})/v_{\text{sh}}}, & \text{if } t \leq t_{\text{diff,c}}, \\ \frac{E_{\text{int}}}{t_{\text{diff,c}} + (r_{\text{out}} - r_{\text{SBO}})/v_{\text{sh}}} e^{-[(t - t_{\text{diff,c}})^2 / t_{\text{diff,c}}^2] / 2}, & \text{if } t > t_{\text{diff,c}}, \end{cases} \quad (\text{S5})$$

where $(r_{\text{out}} - r_{\text{SBO}})/v_{\text{sh}}$ is the diffusion timescale for the breakout shell, and the exponential decline describes the rapid fading after most trapped thermal photons in the shocked CSM have diffused out, followed by radiative and adiabatic cooling (83, 84).

Under local thermal equilibrium, the temperature is determined by balancing the radiation energy density (dominated by the luminosity $u \propto L\tau/cr_{\text{sh}}^2$) with the kinetic energy density: $T_{\text{th}} \approx$

$(u/a)^{1/4}$, where a is the radiation constant. When the shock velocity $v_{\text{sh}}/c \gtrsim 0.05$ occurs, there is not enough time to form a local photon-electron thermal equilibrium, and blackbody radiation cannot build up (34, 32). In such a case, photons and electrons are in Compton equilibrium at a higher temperature T . One can define a thermal coupling coefficient, η , as the ratio of the photon density required for thermal equilibrium to the number of photons produced by free-free emission over the available photon-production timescale in the breakout shell, thereby assessing whether local thermal equilibrium can be established (34, 32). If $\eta \lesssim 1$, the radiation is in local thermal equilibrium and the observed temperature is $T_{\text{obs}} = T_{\text{th}}$; while if $\eta \gtrsim 1$, T_{obs} can be modified by Comptonization as $T_{\text{obs}} = T_{\text{th}}\eta^2/\xi(T_{\text{obs}})^2$, where $\xi(T_{\text{obs}})$ represents photons produced at lower frequencies and Comptonized to a temperature of T . In our case, T_{obs} at the early cooling phase can exceed the thermal equilibrium temperature T_{th} by a factor of ~ 3 –4, suggesting that the emission is not in full local thermal equilibrium but remains partially thermalized. We therefore define a color correction factor, $f_{\text{col}} = T_{\text{obs}}/T_{\text{th}}$, to account for the deviation of the source temperature from the blackbody temperature expected under complete thermal equilibrium, which is used in the main text. The X-ray spectrum is expected to show a Wien peak and is conventionally approximated as a diluted blackbody spectrum characterized by T_{obs} when modeling the X-ray light curve, an assumption that is also supported by the observed spectral properties of EP260321a.

The observed light curve and temperature are modified by the geometric light-travel-time effect, whereby photons emitted simultaneously from different locations on a spherical surface reach the observer at different times. Then, we use this model to simultaneously fit the X-ray light curve and temperature of EP260321a with a total of five free parameters: outer boundary r_{out} , ratio between inner and outer boundary $r_{\text{in}}/r_{\text{out}}$, shock dimensionless velocity $\beta_{\text{sh}} = v_{\text{sh}}/c$, CSM mass M_{CSM} , and time of shock breakout relative to the first observed data point t_{first} . We perform Markov Chain Monte Carlo fitting with *emcee* to model the X-ray light curve and temperature of EP260321a, with the resulting fits shown in Figure 1. The median values with 1σ regions of the key parameters are $r_{\text{out}} = 324.5^{+2.7}_{-3.6} R_{\odot}$, $r_{\text{in}}/r_{\text{out}} = 0.807^{+0.003}_{-0.003}$ (corresponding to $r_{\text{in}} = 261.7^{+2.2}_{-2.7} R_{\odot}$), $\beta_{\text{sh}} = 0.0894^{+0.0003}_{-0.0003}$, $\log_{10}(M_{\text{CSM}}/M_{\odot}) = -3.88^{+0.01}_{-0.01}$, and $t_{\text{first}} = 142.1^{+5.4}_{-8.8}$ s.

Supernova light-curve modeling

The fitting results presented in Figure 4 suggest that the traditional radioactive model can provide a plausible explanation for SN 2026gzf, but a high nickel-to-ejecta mass ratio and extreme ^{56}Ni mixing are required, which somewhat reduces the feasibility of the model. Therefore, here we try to model the supernova in alternative scenarios — in particular, by invoking other energy sources. As usually suggested for unusual supernova phenomena (85, 86, 87), a powerful central engine (e.g., a millisecond magnetar) could be formed in SN 2026gzf, injecting energy into the supernova ejecta during its spin-down. In our calculations, as well for the radioactive model, the monochromatic luminosities in different filters are obtained by considering blackbody spectra for the supernova emission. However, for short-wavelength emission (i.e., Mephisto- ν , Mephisto- u , and u), an additional phenomenological suppression factor such as $f_{\text{UV}} = [\min(1, \lambda_{\text{rest}}/4500 \text{ \AA})]^3$ is invoked to match the observational data (88, 89), where λ_{rest} is the rest-frame wavelength of each filter. The fitting results with a magnetar engine are shown in Extended Data Figure S3. As shown, the shallow increase of SN 2026gzf has been well explained, because the t^{-2} behavior of the energy release from the magnetar naturally determines a logarithmic increase (86). Moreover, the inferred parameter values are also consistent with previous work for other SNe Ic-BL in the magnetar engine model (90, 30, 87). Nevertheless, in order to account for the tail of the supernova emission, $0.26 M_{\odot}$ of ^{56}Ni is still required.

In addition, owing to the potentially complicated CSM environment, one may suggest that SN 2026gzf is powered by an interaction of the supernova ejecta with the CSM. However, this model inevitably requires the presence of a highly optically thick CSM shell, which would determine a large optical depth for soft X-rays due to photoionization or electron scattering, and thus can in principle obscure the previous SBO X-ray emission. In principle, the potential anisotropy of the supernova explosion could still help circumvent this difficulty. The more serious challenge to the CSM-interaction model is that it is actually difficult for the nearly static CSM to generate sufficiently broad lines to match the spectral observations of SN 2026gzf.

The optical data at $\lesssim 10^4$ s always exceed the supernova early emission, regardless of the power scenario. The empirical blackbody fit of the early data gives a blackbody radius $\sim 5 \times 10^{14}$ cm. This makes it difficult to explain with the supernova emission as well as the cooling of the SBO, no

matter which energy mechanism is invoked. Alternatively, this early optical bump is likely to arise from another emission region, which is of a transrelativistic velocity. Here, we tentatively describe the early optical emission by using an evolving blackbody component parameterized as

$$T_{\text{BB}} = T_0 \left(\frac{\hat{t}}{t_{\text{ref}}} \right)^{-\alpha_T}, \quad u(\hat{t}) = \Gamma\beta = u_0 \left(\frac{\hat{t}}{t_{\text{ref}}} \right)^{-\alpha_u}, \quad (\text{S6})$$

where the dynamical time $\hat{t}$ is defined to relate to the observer's time t by

$$\hat{t} = \frac{t}{(1+z)(1-\beta)} + \frac{1-\beta_{\text{ej}}}{1-\beta} \frac{R_{\text{SBO}}}{v_{\text{ej}}}. \quad (\text{S7})$$

The best fit, as presented by the dashed lines in Extended Data Figure S3, gives $u_0 \approx 0.89$, $\alpha_u \approx 0.14$, $T_0 \approx 1.83 \times 10^4 \text{ K}$, and $\alpha_T \approx 0.61$ for $t_{\text{ref}} = 10^4 \text{ s}$, corresponding to $\beta \approx 0.67$ and $\Gamma \approx 1.34$ at $t_{\text{ref}} = 10^4 \text{ s}$. Here, the Mephisto- ν band obviously cannot be included by the blackbody description. This fitting result implies that a subrelativistic cocoon could provide a plausible explanation for the early optical bump, as the bump must not be accounted for by the late cooling of the nonrelativistic SBO. A powerful central engine as suggested above may help to understand the existence of such a subrelativistic cocoon, since a relativistic jet can be driven by the engine even though the jet is finally choked in the progenitor.

Radio observations and constraints

We compiled radio follow-up observations of EP260321a/SN 2026gzf from our collaboration and public GCN Circulars, as summarized in Extended Data Table S5. Times are measured relative to the WXT onset, $T_0 = 2026-03-21 \text{ T}12:16:08 \text{ UTC}$; for observations with reported time ranges, the listed epoch is the midpoint of the interval. The observations cover 0.7–23 GHz from 5.1 to 31.1 d after T_0 . No secure radio counterpart is detected. Upper limits were measured from the local root-mean square (RMS) at the optical position of SN 2026gzf in the final images. MeerKAT observations at 1.28 and 3.0 GHz were reduced with the OXKAT pipeline (91), while ATCA observations were calibrated and imaged with standard CASA procedures. The uGMRT Band-5 public nondetection was reduced with the SPAM pipeline as reported in GCN 44227 (Ref. (92)). e-MERLIN 5 GHz C-band observations, reduced in CASA with the e-MERLIN pipeline, also yielded nondetections at $t_{\text{obs}} = 13.489$ and 30.989 d , with RMS values of 22 and $12 \mu\text{Jy beam}^{-1}$ and corresponding 5σ limits of < 110 and $< 60 \mu\text{Jy beam}^{-1}$, respectively. Public VLA and ATCA constraints are included

from GCN Circulars (93, 94, 95, 96). A low-significance VLA feature reported at ~ 13.6 d is not considered a reliable counterpart after subsequent follow-up/reinspection (94, 95), revealing no significant variability. These limits, as presented in Extended Data Figure S4, can provide a useful constraint on the radio emission arising from interactions of supernova ejecta and, in particular, relativistic jets with CSM.

We further examined the radio constraints by comparing two representative external-shock processes, which arise from the propagation of supernova ejecta or a GRB-like relativistic jet into a wind-like CSM. In both scenarios, the density profile of the wind-like CSM is assumed to be $\rho = Ar^{-2}$ with a normalization $A_* \equiv A/(5 \times 10^{11} \text{ g cm}^{-1})$, where $A_* = 1$ approximately corresponds to $\dot{M} = 10^{-5} M_\odot \text{ yr}^{-1}$ for $v_w = 1000 \text{ km s}^{-1}$. Specifically, for the supernova radio emission, we varied the ejecta velocity over $\beta_{\text{ej}} = v_{\text{ej}}/c \in [0.03 - 0.08]$, together with the microphysical parameters $\log \epsilon_e \in [-1.75, -0.25]$ and $\log \epsilon_B \in [-2, -1]$. For the jet afterglow emission, we scan the initial Lorentz factor $\Gamma_0 \in [10, 300]$, the isotropic-equivalent kinetic energy $\log E_{\text{iso}}/\text{erg} \in [51, 53]$, $\log \epsilon_e \in [-1.25, -0.25]$, and $\log \epsilon_B \in [-4, -0.5]$. For each scenario, 10^4 parameter combinations were sampled. The shaded regions in Extended Data Figure S4 show the 1σ range of the predicted flux density at each time or frequency of observation, while the solid and dashed curves denote the representative curves. To be specific, the representative parameter values are taken as $\Gamma_0 = 100$, $E_{\text{iso}} = 10^{52} \text{ erg}$, $\epsilon_e = 0.1$, and $\epsilon_B = 0.01$ for the jet external shock, which are broadly consistent with GRB afterglow observations (97, 98, 28), while for the supernova external shock, we adopted $\beta_{\text{ej}} = 0.1$, $\epsilon_e = 0.1$, and $\epsilon_B = 0.1$ (Refs. (99, 27, 100)).

The observed radio upper limits are consistent with the low fluxes expected from the ejecta-CSM interaction. In contrast, a standard energetic GRB-like afterglow in a comparable wind environment would generally produce substantially brighter radio emission and can exceed the observational limits over a broad region of the scanned parameter space. These radio non-detections therefore disfavor the presence of a luminous, on-axis, GRB-like relativistic jet associated with EP260321a/SN 2026gzf. However, they should not be interpreted as ruling out all possible relativistic or engine-driven outflows, such as an off-axis jet of a viewing angle $\theta_v \gtrsim 45^\circ$ and, in particular, a deeply choked jet. As a result, a weak sub-relativistic cocoon could appear on the line of sight, which contributes to radio emission of flux (101) of $F_\nu \approx 1.8 \times 10^{-5} \text{ mJy}(\beta/0.6)^{31/4}(t_{\text{obs}}/10 \text{ day})^3(\nu/10 \text{ GHz})^{-3/4}$, where $p = 2.5$, $\epsilon_e = 0.1$, and $\epsilon_B = 0.01$.

are adopted and a typical density of $n = 1\text{cm}^{-3}$ is used because the cocoon has moved into the interstellar medium rather than staying in the wind-like CSM. Obviously, this flux is safely below the current upper limits for fiducial parameters.

Event-rate density

We estimate the local event-rate density of X-ray shock breakout $\rho_{0,\text{SBO}}$ from the single detection of EP260321a in the systematic WXT survey, using the V_{max} method (102) as implemented for systematic study of high-energy transients (103, 104). For EP260321a at $z = 0.034$ (luminosity distance $D_L \approx 150$ Mpc), we compute the maximum volume within which the source would have been detectable given its luminosity light curve and the WXT's sensitivity limit. The field of view of WXT (Ω_{WXT}) is 3600 square degrees. The total exposure time (T_{WXT}) of WXT by March 2026 is ~ 34 million seconds, indicating a survey efficiency of $\sim 50\%$. The WXT simulation indicates that an EP260321a-like event can be well detected by WXT up to a luminosity distance of 250 Mpc with signal-to-noise ratio above 7. We obtain an effective volume-time $\mathcal{M} = V_{\text{max}}T_{\text{WXT}} \approx 0.006$ Gpc³ yr, where V_{max} is the monitored maximum volume weighted by the density evolution and time dilation,

$$V_{\text{max}} = \int_0^{z_{\text{max}}} \frac{\Omega_{\text{WXT}}}{4\pi} \frac{f(z)}{(1+z)} \frac{dV(z)}{dz} dz. \quad (\text{S8})$$

The comoving volume is given by

$$\frac{dV(z)}{dz} = \frac{c}{H_0} \frac{4\pi D_L(z)^2}{(1+z)^2 [\Omega_M(1+z)^3 + \Omega_\Lambda]^{1/2}}, \quad (\text{S9})$$

where $D_L(z)$ is the luminosity distance at the corresponding redshift z . The $f(z)$ factor is model dependent. By assuming that core-collapse supernovae trace the star-formation history, we adopt the analytical formula from Ref. (105),

$$f(z) = \left[(1+z)^{3.4\eta} + \left(\frac{1+z}{5000} \right)^{-0.3\eta} + \left(\frac{1+z}{9} \right)^{-3.5\eta} \right]^{\frac{1}{\eta}}, \quad (\text{S10})$$

where $\eta = -10$.

With a single event detected by WXT, the event-rate density of X-ray shock breakout is

$$\rho_{0,\text{SBO}} \geq \frac{1}{\mathcal{M}} \approx 170_{-140}^{+390} \text{ Gpc}^{-3} \text{ yr}^{-1}, \quad (\text{S11})$$

The error represents 1σ uncertainty, which is calculated from small-sample statistics (106). This value should be regarded as a lower limit on the event-rate density owing to the incompleteness of the identification and classification of EP transients, possibly due to the lack of timely follow-up observations of some EP transients.

For comparison, following the estimate of XRO 080109 (8) and extending the *Swift* operation time to date, we re-estimate the event-rate density from the serendipitous detection of XRO 080109 to be $1.0^{+2.3}_{-0.8} \times 10^4 \text{ Gpc}^{-3} \text{ yr}^{-1}$. The systematic WXT survey yields a lower rate density from EP260321a, yet with an order of magnitude higher luminosity. Both estimates are based on single events and thus remain sensitive to Poisson fluctuations. The higher luminosity of EP260321a compared to XRO 080109, together with their different rate densities, also implies a luminosity-dependent evolution of the SBO rate. To explore this possibility, we make use of EP260321a, together with XRO 080109 and XRF 060218, to probe the X-ray SBO luminosity function following the method in Refs. (103, 104). The luminosity function $\Phi(L)$ is defined as the event-rate density per logarithmic luminosity interval (in units of $\text{Gpc}^{-3} \text{ yr}^{-1} \text{ dex}^{-1}$). A bin width of $\Delta \log L = 1 \text{ dex}$ is adopted in the calculation. Assuming a power-law form for the luminosity function, these three sources are described by a single power-law distribution with a slope of $1.3 \pm 0.4 (1\sigma)$, as shown in the right panel of Figure 3.

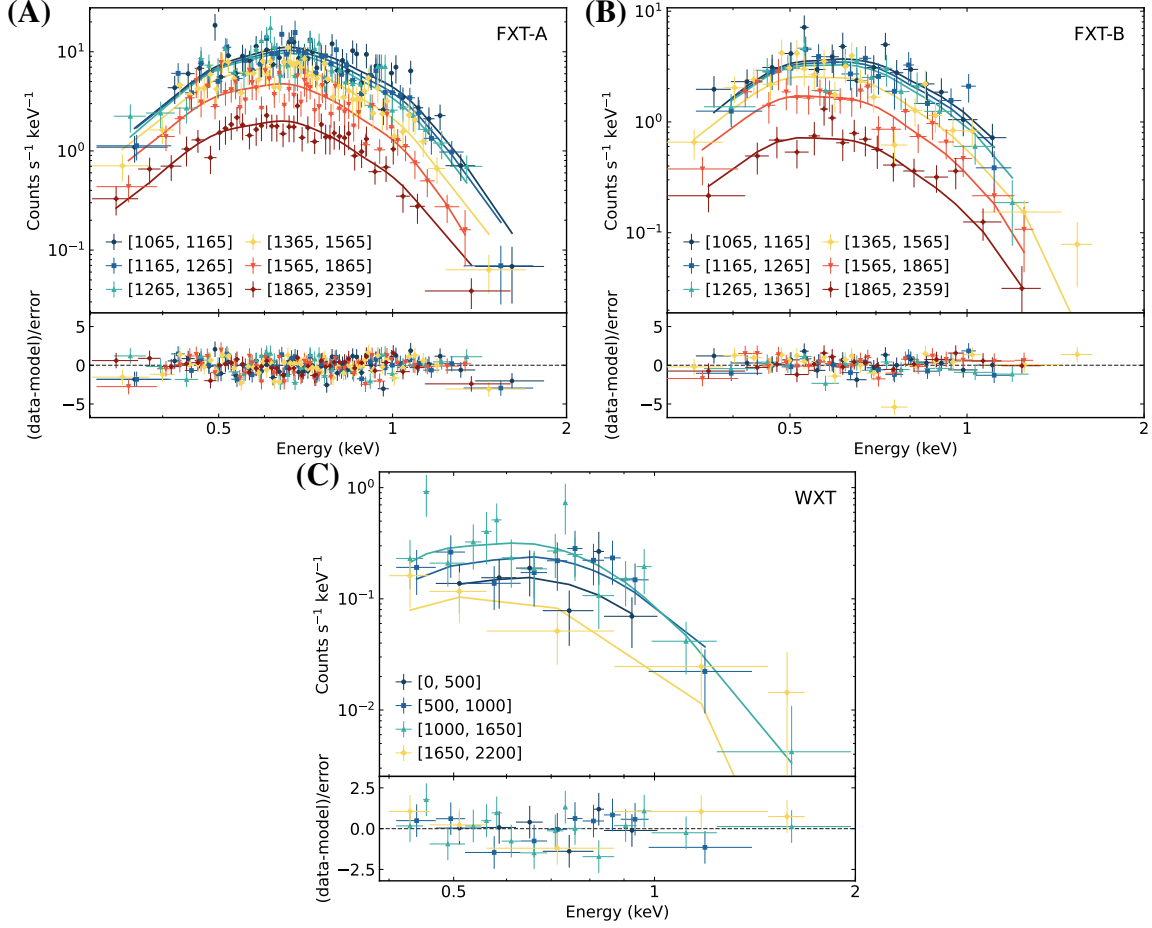


Figure S1: The evolution of the time-resolved WXT and FXT spectra. The data points are shown with 1σ uncertainty and the lines represent the best-fit absorbed blackbody model. The data points of WXT (A) and FXT (B, C) spectra are rebinned to 2σ and 3σ significance, respectively, for clarity. The numbers in the square brackets of the legend represent the time intervals (in seconds) relative to $T_0 = 2026-03-21T12:16:08$ UTC. The best-fit spectral parameters are shown in Extended Table S2.

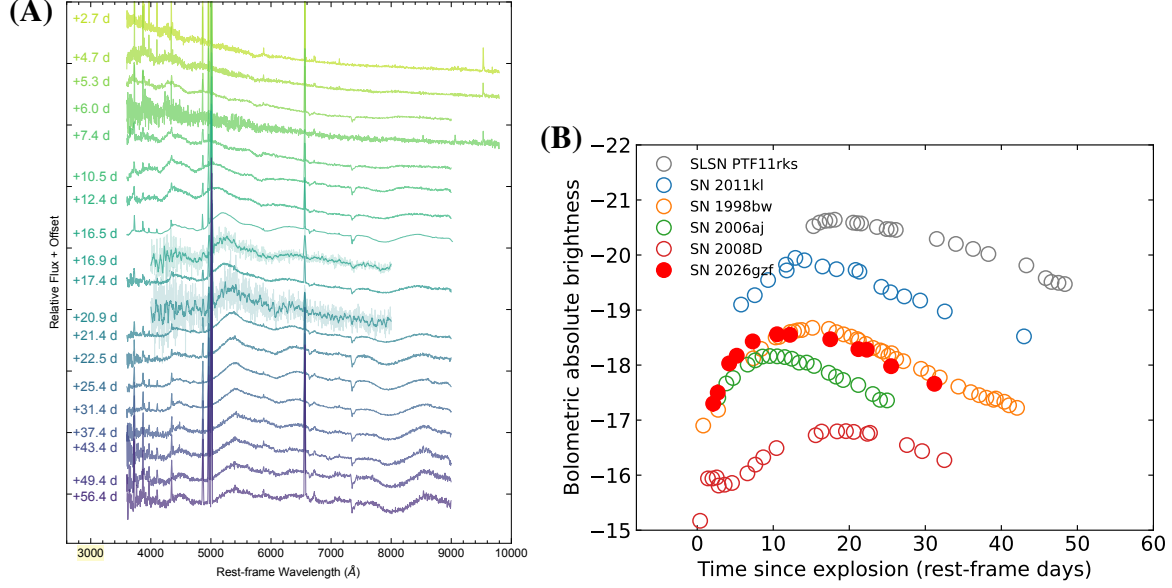


Figure S2: The spectral evolution and bolometric light curve of SN 2026gzf. (A) The Spectra of SN 2026gzf from +2.7 d to +56.4 d. (B) Comparison of the bolometric light curve of SN 2026gzf with those of representative SNe Ic-BL and superluminous SNe. The spectrum-integrated absolute bolometric magnitude of SN 2026gzf is represented by solid circles, plotted alongside the quasibolometric counterparts of representative normal-luminosity, luminous, and superluminous stripped-envelope supernovae. Relative to the other two supernovae with reported X-ray SBO detections, SN 2026gzf rises rapidly to a peak magnitude of $M_{\text{bol}} \approx -18.6$ mag, peaking brighter and slightly later than SN 2006aj, but ~ 10 days earlier than the normal-luminosity SN 2008D. The overall shape of the light curve is similar to that of the well-modelled prototypical broad-lined supernova SN 1998bw, although it peaks earlier, suggesting a somewhat smaller M_{ej}^3/E value. These properties place SN 2026gzf among rapidly evolving luminous supernovae, rather than superluminous ones like PTF11rks.

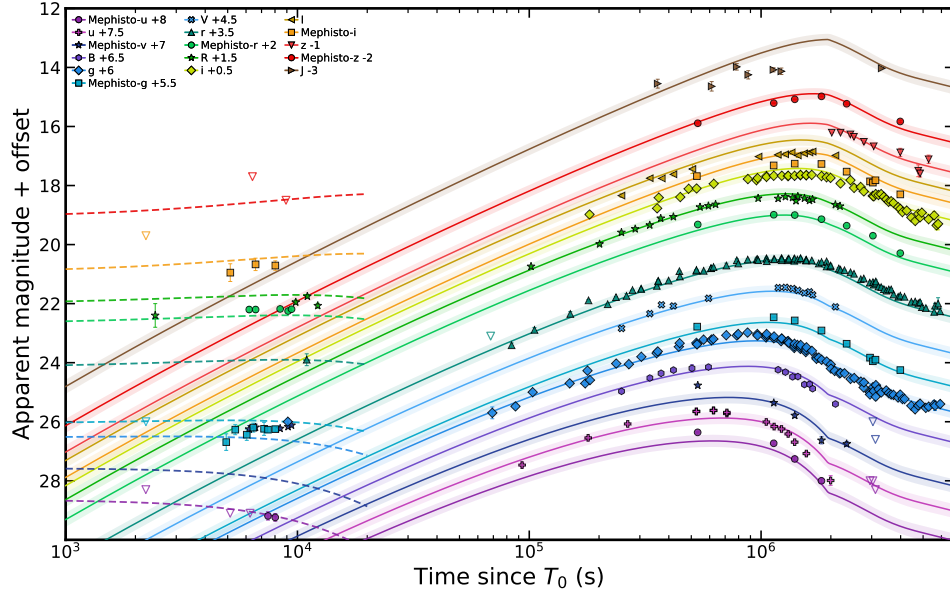


Figure S3: Fits of SN 2026gzf with a magnetar engine. The solid lines give the theoretical light curves from the magnetar engine model for following parameters: initial spin period $P_i = 1.60^{+0.09}_{-0.08}$ ms, dipolar magnetic field strength $B_p = 4.39^{+0.21}_{-0.18} \times 10^{15}$ G, ejecta mass $2.38^{+0.20}_{-0.20} M_\odot$, ^{56}Ni mass $0.26^{+0.01}_{-0.01} M_\odot$, and kinetic energy $2.48^{+1.00}_{-0.97} \times 10^{51}$ erg. The dashed light curves are obtained by assuming blackbody emission with an empirical power-law-evolving temperature and photospheric radius.

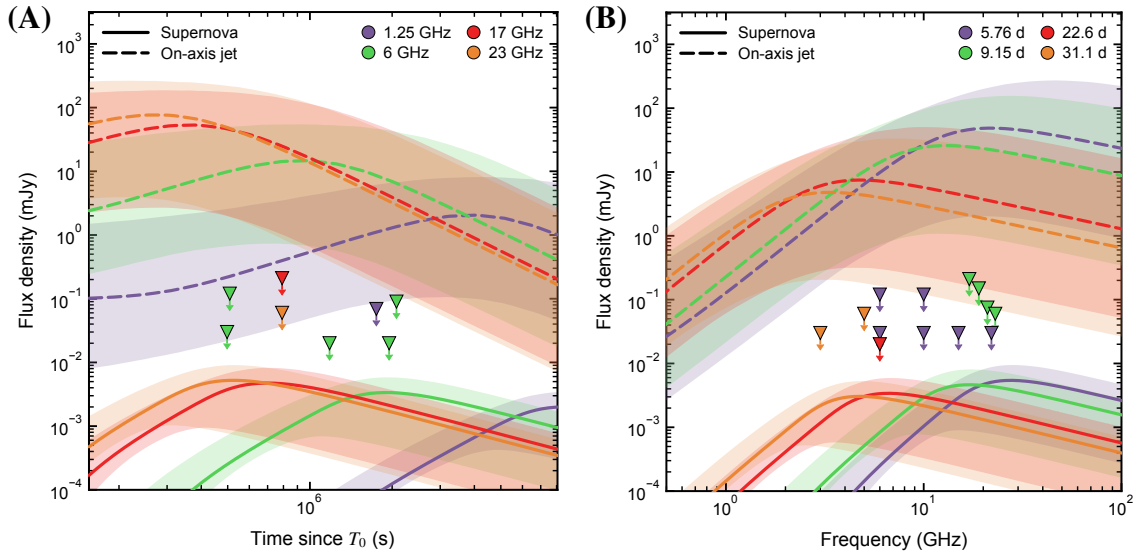


Figure S4: Radio upper limits (downward triangles) compared with model predictions. (A) Radio light curves at 1.25, 6, 17, and 23 GHz, shown as a function of time since T_0 . Solid curves illustrate the expected emission from supernova ejecta–CSM interaction, while dashed curves indicate the radio afterglow of on-axis jets. The shaded regions arise from the typical uncertainties of the model parameters. **(B)** Theoretical radio spectra at representative observing epochs of 5.76, 9.15, 22.6, and 31.1 days.

Table S1: FXT monitoring observation list.

ObsID	Start Time (UTC)	End Time (UTC)	Exp. Time (s)	Upper Limit ²
06800001291 ¹	2026-03-21T21:22:38	2026-03-21T22:58:30	2974	< 4.4
06800001293	2026-03-22T06:55:06	2026-03-22T11:45:26	8321	< 2.9
06800001297	2026-03-23T18:05:31	2026-03-23T22:54:25	8894	< 1.5
06800001298	2026-03-24T10:05:25	2026-03-24T14:52:59	8849	< 1.3
06800001346	2026-04-09T14:15:17	2026-04-09T15:51:08	2962	< 3.7
06800001347	2026-04-10T06:09:48	2026-04-10T09:25:33	4977	< 1.8

¹ The FXT-A was working in the partial-window mode with the thin filter in ObsID 06800001291. Other observations were performed in the full-frame mode with the thin filter.

² The FXT-A/B combined 3σ upper limits (0.4–2.0 keV) are in units of 10^{-14} erg cm⁻² s⁻¹.

Table S2: Spectral parameters of the time-resolved WXT and FXT spectra. All errors represent the 1σ uncertainties.

Time interval (s) ¹	kT_{obs} (eV)	Unabsorbed flux ²	W-stat/dof
WXT segments			
[0, 500]	132^{+30}_{-21}	$1.19^{+0.28}_{-0.23}$	6.0/9
[500, 1000]	136^{+14}_{-12}	$1.85^{+0.32}_{-0.28}$	18.9/19
[1000, 1650]	125^{+10}_{-9}	$1.07^{+0.16}_{-0.14}$	38.6/29
[1650, 2200]	114^{+26}_{-22}	$0.35^{+0.10}_{-0.08}$	14.3/11
FXT-A/B segments			
[1065, 1165]	$122.4^{+3.4}_{-3.3}$	$1.40^{+0.14}_{-0.12}$	132.0/152
[1165, 1265]	$119.6^{+3.4}_{-3.3}$	$1.32^{+0.13}_{-0.11}$	133.8/139
[1265, 1365]	$117.1^{+3.5}_{-3.4}$	$1.25^{+0.13}_{-0.11}$	112.1/133
[1365, 1565]	$110.5^{+2.8}_{-2.8}$	$0.96^{+0.10}_{-0.09}$	189.0/165
[1565, 1865]	$107.8^{+2.9}_{-2.8}$	$0.64^{+0.07}_{-0.06}$	167.1/168
[1865, 2359]	$107.9^{+3.2}_{-3.0}$	$0.27^{+0.03}_{-0.03}$	105.8/138

¹ Time intervals are relative to $T_0 = 2026-03-21T12:16:08$ UTC.

² Flux values (0.4–2.0 keV) and their 1σ errors are given in units of 10^{-10} erg cm⁻² s⁻¹.

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
1.078	u	19.91	0.03	AB	WFST
2.078	u	19.03	0.02	AB	WFST
3.077	u	18.56	0.02	AB	WFST
6.089	u	18.13	0.02	AB	WFST
6.089	u	18.15	0.02	AB	WFST
7.217	u	18.11	0.02	AB	WFST
7.218	u	18.11	0.02	AB	WFST
8.239	u	18.21	0.04	AB	WFST
8.240	u	18.18	0.04	AB	WFST
12.190	u	18.46	0.03	AB	WFST
13.179	u	18.64	0.03	AB	WFST
14.181	u	18.70	0.03	AB	WFST
15.153	u	18.88	0.04	AB	WFST
16.157	u	19.15	0.03	AB	WFST
17.158	u	19.17	0.04	AB	WFST
18.151	u	19.50	0.03	AB	WFST
22.122	u	20.05	0.05	AB	WFST
23.094	u	20.32	0.07	AB	WFST
24.129	u	20.45	0.12	AB	WFST
25.129	u	20.54	0.09	AB	WFST
26.167	u	20.73	0.07	AB	WFST
29.186	u	21.24	0.14	AB	WFST
0.105	g	20.00	0.10	AB	ALT100B
1.115	g	19.55	0.11	AB	WFST
1.729	g	18.70	0.02	AB	TRT-CTO

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
2.064	g	18.27	0.03	AB	ALT100B
2.115	g	18.51	0.05	AB	WFST
2.746	g	18.03	0.02	AB	TRT-CTO
3.114	g	18.00	0.03	AB	WFST
3.523	g	17.80	0.01	AB	TRT-CTO
4.100	g	17.68	0.02	AB	XL-80
4.146	g	17.62	0.01	AB	ALT100B
5.114	g	17.46	0.01	AB	ALT100B
5.157	g	17.34	0.01	AB	XL-80
6.084	g	17.33	0.01	AB	WHUT
6.126	g	17.33	0.02	AB	WFST
6.126	g	17.35	0.02	AB	WFST
6.537	g	17.22	0.02	AB	TRT-CTO
7.070	g	17.19	0.02	AB	WFST
7.070	g	17.21	0.02	AB	WFST
7.206	g	17.20	0.03	AB	ALT100B
8.191	g	17.07	0.03	AB	ALT100B
9.093	g	17.08	0.05	AB	XL-80
9.214	g	17.03	0.03	AB	ALT100B
10.121	g	16.98	0.02	AB	ALT100B
11.143	g	17.03	0.03	AB	ALT100B
11.166	g	17.06	0.02	AB	WFST
12.162	g	17.15	0.01	AB	WFST
12.201	g	17.09	0.01	AB	WHUT
12.234	g	17.03	0.03	AB	ALT100B

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
12.162	g	17.13	0.01	AB	WFST
13.167	g	17.11	0.01	AB	ALT100B
13.141	g	17.15	0.01	AB	WFST
14.121	g	17.24	0.02	AB	WFST
14.192	g	17.19	0.02	AB	ALT100B
14.209	g	17.18	0.01	AB	WHUT
15.041	g	17.28	0.01	AB	XL-80
15.085	g	17.28	0.02	AB	WFST
15.144	g	17.27	0.01	AB	WHUT
15.264	g	17.33	0.02	AB	ALT100B
16.083	g	17.50	0.01	AB	XL-80
16.086	g	17.41	0.02	AB	WFST
16.129	g	17.36	0.01	AB	WHUT
16.614	g	17.40	0.02	AB	TRT-CTO
17.086	g	17.50	0.02	AB	WFST
17.102	g	17.46	0.01	AB	WHUT
17.155	g	17.40	0.03	AB	ALT100B
18.050	g	17.61	0.02	AB	WFST
19.051	g	17.69	0.03	AB	WFST
18.107	g	17.57	0.01	AB	WHUT
18.164	g	17.48	0.01	AB	ALT100B
19.178	g	17.60	0.02	AB	ALT100B
20.050	g	17.78	0.03	AB	WFST
20.051	g	17.83	0.03	AB	WFST
20.535	g	17.83	0.01	AB	TRT-CTO

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
21.116	g	17.90	0.02	AB	ALT100B
22.203	g	17.96	0.02	AB	ALT100B
23.112	g	18.06	0.02	AB	ALT100B
23.164	g	18.14	0.04	AB	WFST
24.057	g	18.23	0.04	AB	WFST
25.200	g	18.30	0.04	AB	WFST
25.471	g	18.35	0.01	AB	NOT
26.504	g	18.36	0.01	AB	TRT-CTO
28.156	g	18.55	0.05	AB	ALT100B
29.180	g	18.52	0.05	AB	ALT100B
31.171	g	18.64	0.05	AB	ALT100B
32.063	g	18.74	0.06	AB	WFST
32.129	g	18.67	0.03	AB	ALT100B
34.092	g	18.87	0.07	AB	WFST
34.117	g	18.90	0.07	AB	WFST
34.151	g	18.96	0.08	AB	WFST
35.178	g	18.83	0.09	AB	ALT100B
35.419	g	18.96	0.02	AB	NOT
36.137	g	19.15	0.11	AB	WFST
36.217	g	18.76	0.08	AB	ALT100B
37.168	g	18.90	0.08	AB	ALT100B
38.097	g	18.93	0.08	AB	WFST
38.107	g	19.02	0.08	AB	WFST
38.186	g	19.01	0.08	AB	ALT100B
40.143	g	18.94	0.06	AB	ALT100B

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
41.146	g	19.03	0.06	AB	ALT100B
42.098	g	19.17	0.09	AB	WFST
42.128	g	19.08	0.07	AB	ALT100B
43.132	g	19.09	0.07	AB	ALT100B
43.143	g	19.21	0.09	AB	WFST
45.155	g	19.18	0.08	AB	ALT100B
46.128	g	19.08	0.06	AB	ALT100B
47.134	g	19.23	0.10	AB	ALT100B
48.135	g	19.23	0.03	AB	ALT100B
51.478	g	19.46	0.04	AB	TRT-CTO
53.478	g	19.58	0.04	AB	TRT-CTO
55.134	g	19.45	0.04	AB	ALT100C
56.140	g	19.39	0.05	AB	ALT100C
59.473	g	19.53	0.06	AB	TRT-CTO
61.158	g	19.49	0.10	AB	ALT100C
64.563	g	19.41	0.12	AB	TRT-CTO
65.136	g	19.46	0.16	AB	ALT100B
68.545	g	19.40	0.10	AB	TRT-CTO
0.127	r	20.40	0.20	AB	ALT100B
1.740	r	18.85	0.08	AB	TRT-CTO
2.040	r	18.64	0.06	AB	WFST
2.085	r	18.39	0.03	AB	ALT100B
2.739	r	18.28	0.02	AB	TRT-CTO
3.040	r	18.18	0.05	AB	WFST
3.227	r	18.04	0.01	AB	GMG

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Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
3.515	r	18.00	0.02	AB	FORS2
3.516	r	18.00	0.02	AB	TRT-CTO
4.111	r	17.90	0.02	AB	XL-80
4.167	r	17.74	0.02	AB	ALT100B
4.168	r	17.69	0.03	AB	WFST
4.618	r	17.72	0.01	AB	FORS2
5.142	r	17.40	0.01	AB	ALT100B
5.170	r	17.58	0.01	AB	XL-80
6.085	r	17.36	0.02	AB	WFST
6.092	r	17.36	0.01	AB	WHUT
6.545	r	17.39	0.02	AB	TRT-CTO
7.042	r	17.27	0.02	AB	WFST
7.042	r	17.28	0.02	AB	WFST
7.217	r	17.26	0.03	AB	ALT100B
8.203	r	17.21	0.08	AB	ALT100B
9.097	r	17.03	0.05	AB	XL-80
9.228	r	17.14	0.04	AB	ALT100B
10.137	r	17.01	0.02	AB	ALT100B
11.050	r	17.01	0.05	AB	XL-80
11.158	r	16.99	0.02	AB	ALT100B
11.243	r	17.01	0.02	AB	WFST
12.209	r	16.98	0.01	AB	WHUT
12.217	r	16.97	0.04	AB	ALT100B
12.249	r	17.03	0.02	AB	WFST
13.152	r	16.99	0.01	AB	ALT100B

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
13.213	r	16.98	0.02	AB	WFST
14.181	r	16.96	0.03	AB	ALT100B
14.207	r	17.00	0.01	AB	WFST
14.216	r	16.98	0.01	AB	WHUT
15.045	r	17.03	0.01	AB	XL-80
15.151	r	16.99	0.01	AB	WHUT
15.253	r	16.98	0.03	AB	ALT100B
16.048	r	16.98	0.01	AB	WFST
16.087	r	17.05	0.01	AB	XL-80
16.137	r	16.97	0.01	AB	WHUT
16.609	r	17.01	0.01	AB	TRT-CTO
17.109	r	16.98	0.01	AB	WHUT
17.143	r	16.96	0.02	AB	ALT100B
18.083	r	17.01	0.01	AB	WFST
18.115	r	17.01	0.01	AB	WHUT
18.143	r	17.04	0.01	AB	ALT100B
19.084	r	17.00	0.02	AB	WFST
19.166	r	17.04	0.01	AB	ALT100B
20.527	r	17.14	0.01	AB	TRT-CTO
21.104	r	17.09	0.03	AB	ALT100B
22.191	r	17.07	0.02	AB	ALT100B
22.192	r	17.21	0.02	AB	WFST
23.057	r	17.21	0.02	AB	WFST
23.100	r	17.21	0.05	AB	ALT100B
25.059	r	17.28	0.02	AB	WFST

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
25.474	r	17.35	0.01	AB	NOT
26.496	r	17.37	0.01	AB	TRT-CTO
28.144	r	17.44	0.02	AB	ALT100B
29.134	r	17.49	0.01	AB	ALT100A
31.159	r	17.67	0.02	AB	ALT100B
32.117	r	17.65	0.02	AB	ALT100B
34.089	r	17.75	0.03	AB	WFST
35.159	r	17.77	0.03	AB	ALT100B
35.422	r	17.86	0.01	AB	NOT
36.128	r	17.78	0.04	AB	WFST
36.192	r	17.75	0.05	AB	ALT100B
37.141	r	17.92	0.05	AB	ALT100B
37.160	r	17.84	0.04	AB	WFST
38.070	r	17.96	0.04	AB	WFST
38.080	r	17.94	0.03	AB	WFST
38.159	r	17.94	0.05	AB	ALT100B
39.071	r	18.00	0.04	AB	WFST
39.202	r	18.01	0.07	AB	ALT100B
40.197	r	18.06	0.05	AB	ALT100B
41.074	r	18.06	0.04	AB	WFST
41.099	r	18.07	0.04	AB	WFST
41.172	r	18.02	0.04	AB	ALT100B
42.141	r	18.14	0.04	AB	WFST
42.144	r	18.02	0.04	AB	ALT100B
43.074	r	18.13	0.04	AB	WFST

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
43.146	r	18.19	0.04	AB	ALT100B
44.074	r	18.24	0.06	AB	WFST
45.167	r	18.27	0.05	AB	ALT100B
46.078	r	18.20	0.05	AB	WFST
46.142	r	18.23	0.04	AB	ALT100B
47.104	r	18.24	0.04	AB	WFST
48.077	r	18.29	0.05	AB	WFST
48.121	r	18.27	0.02	AB	ALT100B
50.095	r	18.37	0.05	AB	WFST
51.482	r	18.44	0.03	AB	TRT-CTO
53.486	r	18.43	0.03	AB	TRT-CTO
55.140	r	18.49	0.05	AB	ALT100B
56.161	r	18.45	0.05	AB	ALT100B
57.148	r	18.62	0.04	AB	ALT100B
59.480	r	18.65	0.04	AB	TRT-CTO
64.569	r	18.78	0.10	AB	TRT-CTO
65.121	r	18.57	0.08	AB	ALT100B
67.146	r	18.50	0.20	AB	ALT100B
68.552	r	18.76	0.10	AB	TRT-CTO
2.107	i	18.48	0.10	AB	ALT100B
4.121	i	18.27	0.03	AB	XL-80
4.188	i	17.91	0.05	AB	ALT100B
5.185	i	17.89	0.02	AB	XL-80
5.639	i	17.61	0.03	AB	TRT-CTO
6.099	i	17.60	0.01	AB	WHUT

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
7.233	i	17.44	0.03	AB	ALT100B
10.157	i	17.28	0.04	AB	ALT100B
11.054	i	17.32	0.10	AB	XL-80
11.180	i	17.25	0.05	AB	ALT100B
12.193	i	17.18	0.05	AB	ALT100B
12.216	i	17.21	0.02	AB	WHUT
13.129	i	17.17	0.05	AB	ALT100B
14.224	i	17.17	0.02	AB	WHUT
15.120	i	17.18	0.02	AB	ALT100B
15.159	i	17.17	0.02	AB	WHUT
16.144	i	17.17	0.01	AB	WHUT
17.117	i	17.15	0.02	AB	WHUT
17.126	i	17.14	0.03	AB	ALT100B
18.122	i	17.15	0.02	AB	WHUT
18.126	i	17.15	0.03	AB	ALT100B
19.149	i	17.14	0.02	AB	ALT100B
21.087	i	17.18	0.03	AB	ALT100B
22.174	i	17.22	0.05	AB	ALT100B
23.076	i	17.24	0.03	AB	ALT100B
25.477	i	17.43	0.01	AB	NOT
28.127	i	17.32	0.04	AB	ALT100B
29.151	i	17.58	0.04	AB	ALT100B
31.142	i	17.55	0.03	AB	ALT100B
32.100	i	17.70	0.03	AB	ALT100B
35.117	i	17.85	0.10	AB	ALT100B

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
35.426	i	17.87	0.01	AB	NOT
36.170	i	17.79	0.08	AB	ALT100B
37.113	i	17.82	0.08	AB	ALT100B
38.128	i	17.79	0.07	AB	ALT100B
39.174	i	17.88	0.11	AB	ALT100B
40.170	i	17.97	0.06	AB	ALT100B
41.115	i	18.05	0.05	AB	ALT100B
42.107	i	18.03	0.05	AB	ALT100B
43.113	i	18.09	0.08	AB	ALT100B
45.138	i	18.23	0.07	AB	ALT100B
46.109	i	18.29	0.06	AB	ALT100B
47.114	i	18.40	0.06	AB	ALT100B
49.151	i	18.35	0.05	AB	ALT100B
51.497	i	18.58	0.10	AB	TRT-CTO
53.495	i	18.70	0.10	AB	TRT-CTO
55.118	i	18.46	0.12	AB	ALT100B
56.140	i	18.58	0.10	AB	ALT100B
57.126	i	18.42	0.08	AB	ALT100B
61.125	i	18.64	0.10	AB	ALT100B
64.579	i	18.51	0.13	AB	TRT-CTO
66.145	i	18.85	0.15	AB	ALT100B
67.142	i	18.80	0.20	AB	ALT100B
23.366	z	17.21	0.01	AB	NOT
25.481	z	17.21	0.01	AB	NOT
28.098	z	17.27	0.05	AB	ALT100A

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
29.110	z	17.35	0.04	AB	ALT100A
32.136	z	17.52	0.06	AB	ALT50A
35.429	z	17.67	0.01	AB	NOT
46.131	z	17.88	0.14	AB	ALT50A
55.156	z	18.50	0.20	AB	ALT100C
56.161	z	18.59	0.12	AB	ALT100C
61.138	z	18.11	0.14	AB	ALT100C
0.026	Mephisto-u	>20.3	-	AB	Mephisto
0.059	Mephisto-u	>21.1	-	AB	Mephisto
0.072	Mephisto-u	>21.1	-	AB	Mephisto
0.086	Mephisto-u	21.19	0.14	AB	Mephisto
0.093	Mephisto-u	21.24	0.13	AB	Mephisto
6.164	Mephisto-u	18.36	0.02	AB	Mephisto
13.124	Mephisto-u	18.73	0.05	AB	Mephisto
16.173	Mephisto-u	19.26	0.07	AB	Mephisto
21.084	Mephisto-u	20.00	0.07	AB	Mephisto
34.162	Mephisto-u	>20.0	-	AB	Mephisto
35.146	Mephisto-u	>20.0	-	AB	Mephisto
36.041	Mephisto-u	>20.3	-	AB	Mephisto
0.072	Mephisto-v	19.25	0.12	AB	Mephisto
0.076	Mephisto-v	19.19	0.11	AB	Mephisto
0.097	Mephisto-v	19.23	0.05	AB	Mephisto
0.105	Mephisto-v	19.17	0.04	AB	Mephisto
0.109	Mephisto-v	19.12	0.04	AB	Mephisto
6.172	Mephisto-v	17.77	0.04	AB	Mephisto

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
13.130	Mephisto-v	18.36	0.05	AB	Mephisto
16.175	Mephisto-v	18.79	0.07	AB	Mephisto
21.091	Mephisto-v	19.63	0.12	AB	Mephisto
27.088	Mephisto-v	19.75	0.08	AB	Mephisto
35.156	Mephisto-v	>19.0	-	AB	Mephisto
36.050	Mephisto-v	>19.6	-	AB	Mephisto
0.026	Mephisto-g	>20.5	-	AB	Mephisto
0.057	Mephisto-g	21.19	0.28	AB	Mephisto
0.062	Mephisto-g	20.77	0.17	AB	Mephisto
0.070	Mephisto-g	20.94	0.19	AB	Mephisto
0.075	Mephisto-g	20.70	0.15	AB	Mephisto
0.083	Mephisto-g	20.74	0.22	AB	Mephisto
0.086	Mephisto-g	20.77	0.08	AB	Mephisto
0.093	Mephisto-g	20.75	0.08	AB	Mephisto
6.136	Mephisto-g	17.28	0.01	AB	Mephisto
13.124	Mephisto-g	16.96	0.01	AB	Mephisto
16.173	Mephisto-g	17.07	0.01	AB	Mephisto
21.084	Mephisto-g	17.41	0.01	AB	Mephisto
27.083	Mephisto-g	17.86	0.01	AB	Mephisto
34.162	Mephisto-g	18.33	0.06	AB	Mephisto
35.146	Mephisto-g	18.45	0.06	AB	Mephisto
36.041	Mephisto-g	18.40	0.05	AB	Mephisto
46.124	Mephisto-g	18.75	0.02	AB	Mephisto
0.072	Mephisto-r	20.20	0.10	AB	Mephisto
0.076	Mephisto-r	20.20	0.10	AB	Mephisto

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
0.097	Mephisto-r	20.18	0.06	AB	Mephisto
0.105	Mephisto-r	20.25	0.06	AB	Mephisto
0.109	Mephisto-r	20.19	0.05	AB	Mephisto
6.172	Mephisto-r	17.32	0.01	AB	Mephisto
13.130	Mephisto-r	16.99	0.01	AB	Mephisto
16.175	Mephisto-r	17.00	0.01	AB	Mephisto
21.091	Mephisto-r	17.14	0.01	AB	Mephisto
27.088	Mephisto-r	17.36	0.01	AB	Mephisto
35.156	Mephisto-r	17.70	0.07	AB	Mephisto
46.128	Mephisto-r	18.29	0.01	AB	Mephisto
0.026	Mephisto-i	>19.7	-	AB	Mephisto
0.059	Mephisto-i	20.95	0.30	AB	Mephisto
0.076	Mephisto-i	20.67	0.21	AB	Mephisto
0.093	Mephisto-i	20.71	0.16	AB	Mephisto
6.120	Mephisto-i	17.68	0.01	AB	Mephisto
13.124	Mephisto-i	17.32	0.01	AB	Mephisto
16.173	Mephisto-i	17.26	0.01	AB	Mephisto
21.086	Mephisto-i	17.27	0.01	AB	Mephisto
27.083	Mephisto-i	17.53	0.01	AB	Mephisto
34.162	Mephisto-i	17.86	0.04	AB	Mephisto
35.146	Mephisto-i	17.90	0.05	AB	Mephisto
36.045	Mephisto-i	17.82	0.03	AB	Mephisto
46.124	Mephisto-i	18.30	0.02	AB	Mephisto
0.074	Mephisto-z	>19.7	-	AB	Mephisto
0.103	Mephisto-z	>20.5	-	AB	Mephisto

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
6.172	Mephisto-z	17.89	0.02	AB	Mephisto
13.130	Mephisto-z	17.21	0.01	AB	Mephisto
16.175	Mephisto-z	17.08	0.01	AB	Mephisto
21.091	Mephisto-z	16.98	0.01	AB	Mephisto
27.088	Mephisto-z	17.23	0.03	AB	Mephisto
46.128	Mephisto-z	17.83	0.04	AB	Mephisto
2.894	B	18.47	0.05	Vega	KAIT
3.834	B	18.02	0.05	Vega	KAIT
4.288	B	17.86	0.01	Vega	ZTSh
4.848	B	17.75	0.04	Vega	KAIT
5.826	B	17.69	0.12	Vega	KAIT
6.864	B	17.65	0.04	Vega	KAIT
13.711	B	17.74	0.04	Vega	KAIT
14.741	B	17.82	0.04	Vega	KAIT
15.828	B	17.98	0.04	Vega	KAIT
16.685	B	17.97	0.05	Vega	KAIT
17.809	B	18.24	0.06	Vega	KAIT
18.732	B	18.23	0.07	Vega	KAIT
19.315	B	18.38	0.01	Vega	Zeiss-1000
24.227	B	18.90	0.01	Vega	Zeiss-1000
2.895	V	18.34	0.03	Vega	KAIT
3.834	V	17.84	0.03	Vega	KAIT
4.303	V	17.54	0.01	Vega	ZTSh
4.844	V	17.58	0.03	Vega	KAIT
6.864	V	17.32	0.03	Vega	KAIT

Continued on next page

Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
13.712	V	16.96	0.02	Vega	KAIT
14.742	V	16.95	0.01	Vega	KAIT
15.829	V	16.99	0.01	Vega	KAIT
16.686	V	17.05	0.02	Vega	KAIT
17.810	V	17.14	0.02	Vega	KAIT
18.732	V	17.10	0.02	Vega	KAIT
19.317	V	17.20	0.01	Vega	Zeiss-1000
24.231	V	17.61	0.01	Vega	Zeiss-1000
0.028	R	20.90	0.40	Vega	TRT-SBO
0.114	R	20.45	0.07	Vega	Mondy
0.128	R	20.25	0.06	Vega	Mondy
0.142	R	20.57	0.08	Vega	Mondy
1.178	R	19.24	0.01	Vega	Mondy
2.327	R	18.48	0.05	Vega	Astrosib-500
2.891	R	18.09	0.02	Vega	KAIT
3.316	R	17.97	0.01	Vega	Astrosib-500
3.831	R	17.84	0.02	Vega	KAIT
4.271	R	17.61	0.01	Vega	ZTSh
4.845	R	17.56	0.02	Vega	KAIT
6.311	R	17.23	0.01	Vega	MTM-500
6.861	R	17.19	0.02	Vega	KAIT
7.366	R	17.14	0.02	Vega	Zeiss-1000
11.239	R	16.93	0.01	Vega	MTM-500
13.713	R	16.94	0.01	Vega	KAIT
14.738	R	16.88	0.01	Vega	KAIT

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Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
15.830	R	16.91	0.01	Vega	KAIT
16.374	R	17.02	0.04	Vega	Astrosib-500
16.682	R	16.87	0.01	Vega	KAIT
17.811	R	16.93	0.02	Vega	KAIT
18.729	R	17.01	0.02	Vega	KAIT
19.306	R	16.97	0.01	Vega	Zeiss-1000
24.214	R	17.15	0.01	Vega	Zeiss-1000
25.353	R	17.20	0.01	Vega	Zeiss-1000
2.892	I	18.34	0.06	Vega	KAIT
3.832	I	17.75	0.04	Vega	KAIT
4.317	I	17.76	0.01	Vega	ZTSh
4.846	I	17.62	0.05	Vega	KAIT
5.851	I	17.44	0.15	Vega	KAIT
11.288	I	17.02	0.02	Vega	MTM-500
13.674	I	16.96	0.05	Vega	KAIT
14.705	I	16.94	0.06	Vega	KAIT
15.695	I	16.88	0.06	Vega	KAIT
16.675	I	16.98	0.07	Vega	KAIT
17.742	I	16.91	0.06	Vega	KAIT
18.715	I	16.88	0.06	Vega	KAIT
19.312	I	16.85	0.01	Vega	Zeiss-1000
24.223	I	16.98	0.01	Vega	Zeiss-1000
4.146	J	17.55	0.15	Vega	ALT100C
7.087	J	17.64	0.16	Vega	ALT100C
9.091	J	16.98	0.11	Vega	ALT100C

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Table S3: Photometric Table

Time (day)	Band	Magnitude	Error	System	Telescope
10.191	J	17.25	0.14	Vega	ALT100C
13.093	J	17.08	0.07	Vega	ALT100C
14.159	J	17.13	0.11	Vega	ALT100C
38.369	J	17.03	0.01	Vega	NOT
0.713	clear	20.04	0.07	Vega	KAIT
0.862	clear	19.57	0.06	Vega	KAIT
2.893	clear	18.35	0.01	Vega	KAIT
3.833	clear	18.19	0.01	Vega	KAIT
3.861	clear	17.78	0.01	Vega	KAIT
4.847	clear	17.71	0.01	Vega	KAIT
4.875	clear	17.30	0.02	Vega	KAIT
5.852	clear	17.37	0.03	Vega	KAIT
6.863	clear	17.20	0.01	Vega	KAIT
13.714	clear	16.89	0.01	Vega	KAIT
14.740	clear	17.06	0.01	Vega	KAIT
15.818	clear	17.08	0.01	Vega	KAIT
15.827	clear	16.94	0.01	Vega	KAIT
16.684	clear	16.98	0.01	Vega	KAIT
17.803	clear	16.90	0.01	Vega	KAIT
17.813	clear	17.02	0.01	Vega	KAIT
18.731	clear	17.09	0.01	Vega	KAIT
18.813	clear	17.12	0.01	Vega	KAIT

Table S4: Log of spectroscopy of SN 2026gzf.

UTC Date	Δt (day)	Range (Å)	Airmass	Total Exp. (s)	Telescope	Instrument
Mar. 24	2.7	3800–10000	1.4	6000	Shane	Kast
Mar. 26	4.7	3800–10000	1.0	3600	Shane	Kast
Mar. 26	5.3	3800–9200	1.3	1800	NOT	ALFOSC
Mar. 27	6.0	3800–10000	1.3	3600	Shane	Kast
Mar. 28	7.4	3800–9200	1.2	2400	NOT	ALFOSC
Mar. 31	10.5	3800–9200	1.3	2400	NOT	ALFOSC
Apr. 02	12.4	3800–9200	1.1	2400	NOT	ALFOSC
Apr. 06	16.5	3410–9330	1.2	480×8	VLT	FORS2
Apr. 07	16.9	4000–9000	1.6	3600	Xinglong 2.16 m	BFOSC
Apr. 07	17.4	3800–9200	1.1	2400	NOT	ALFOSC
Apr. 11	20.9	4000–9000	1.6	3600	Xinglong 2.16 m	BFOSC
Apr. 11	21.4	3800–9200	1.1	2400	NOT	ALFOSC
Apr. 12	22.5	3800–9200	1.9	2400	NOT	ALFOSC
Apr. 15	25.4	3800–9200	1.2	2400	NOT	ALFOSC
Apr. 21	31.4	3800–9200	1.1	2400	NOT	ALFOSC
Apr. 27	37.4	3800–9200	1.2	2400	NOT	ALFOSC
May. 03	43.4	3800–9200	1.3	2400	NOT	ALFOSC
May. 09	49.4	3800–9200	1.3	2400	NOT	ALFOSC
May. 16	56.4	3800–9200	1.3	2400	NOT	ALFOSC

Table S5: Radio observations of EP260321a/SN 2026gzf. Epochs are computed relative to $T_0 = 2026-03-21T12:16:08$ UTC. For observations with reported UTC ranges, the listed epoch is the midpoint. Upper limits are 3σ unless otherwise stated.

Facility	Epoch / UTC range	Δt (d)	ν (GHz)	Result	Obs. code / ref.
uGMRT	2026-03-26 15:00–16:00	5.135	0.7	rms = $120 \mu\text{Jy}$; $< 360 \mu\text{Jy}$	49_076/49_114; GCN 44227 (92), this work
uGMRT	2026-03-26 18:00–19:00	5.260	1.1	rms = $90 \mu\text{Jy}$; $< 270 \mu\text{Jy}$	49_076/49_114; GCN 44227 (92), this work
VLA	2026-03-27 05:12–07:42	5.758	6, 10, 15, 22	$< 30 \mu\text{Jy beam}^{-1}$ at each band	GCN 44229 (93)
ATCA	2026-03-27 09:00–15:00	5.889	6.0, 10.0	rms $\simeq 40 \mu\text{Jy}$; $< 120 \mu\text{Jy}$	CX595; this work
ATCA	2026-03-30 10:30–21:30	9.155	17, 19, 21, 23	$< 210, 150, 75, 60 \mu\text{Jy}$ at 17, 19, 21, 23 GHz	GCN 44403 (96)
ATCA	2026-04-02 10:30–14:00	11.999	20.0	rms $\simeq 32 \mu\text{Jy}$; $< 96 \mu\text{Jy}$	C3735; this work
e-MERLIN	2026-04-01 04:05 – 04-02 02:04 & 2026-04-06 19:25 – 04-07 07:30	13.489	5.0	rms = $22 \mu\text{Jy}$; $5\sigma < 110 \mu\text{Jy}$	CY21232; this work
VLA	2026-04-04 04:03–05:03	13.658	6.0	low-significance feature; rms = $5 \mu\text{Jy}$; $4\sigma \lesssim 20 \mu\text{Jy}$	GCN 44239 (94)
MeerKAT	2026-04-10 18:17–19:04	20.267	1.28	rms $\simeq 23 \mu\text{Jy}$; $< 69 \mu\text{Jy}$	DDT-20260331; this work
VLA	2026-04-13 01:20–02:20	22.565	6.0	no confirmed counterpart; rms = $5 \mu\text{Jy}$; $4\sigma \lesssim 20 \mu\text{Jy}$	GCN 44357 (95)
MeerKAT	2026-04-13 16:02–16:49	23.173	S41 3.0	rms $\simeq 11 \mu\text{Jy}$; $< 33 \mu\text{Jy}$	DDT-20260331;